INCH-POUND
MIL-DTL-83507/15A
25 July 2025
SUPERSEDING
MIL-T-83507/15
7 December 1977

DETAIL SPECIFICATION SHEET

TOOL KITS, ELECTRICAL CONNECTORS, CONTACTS, AND CONNECTOR AND CABLE ACCESSORIES: TOOL KIT, GENERAL PURPOSE, BACKSHELL HARDWARE TOOLS

Inactive for new design after September 1, 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the tool kit described herein shall consist of this document and the latest issue of Specification MIL-DTL-83507.

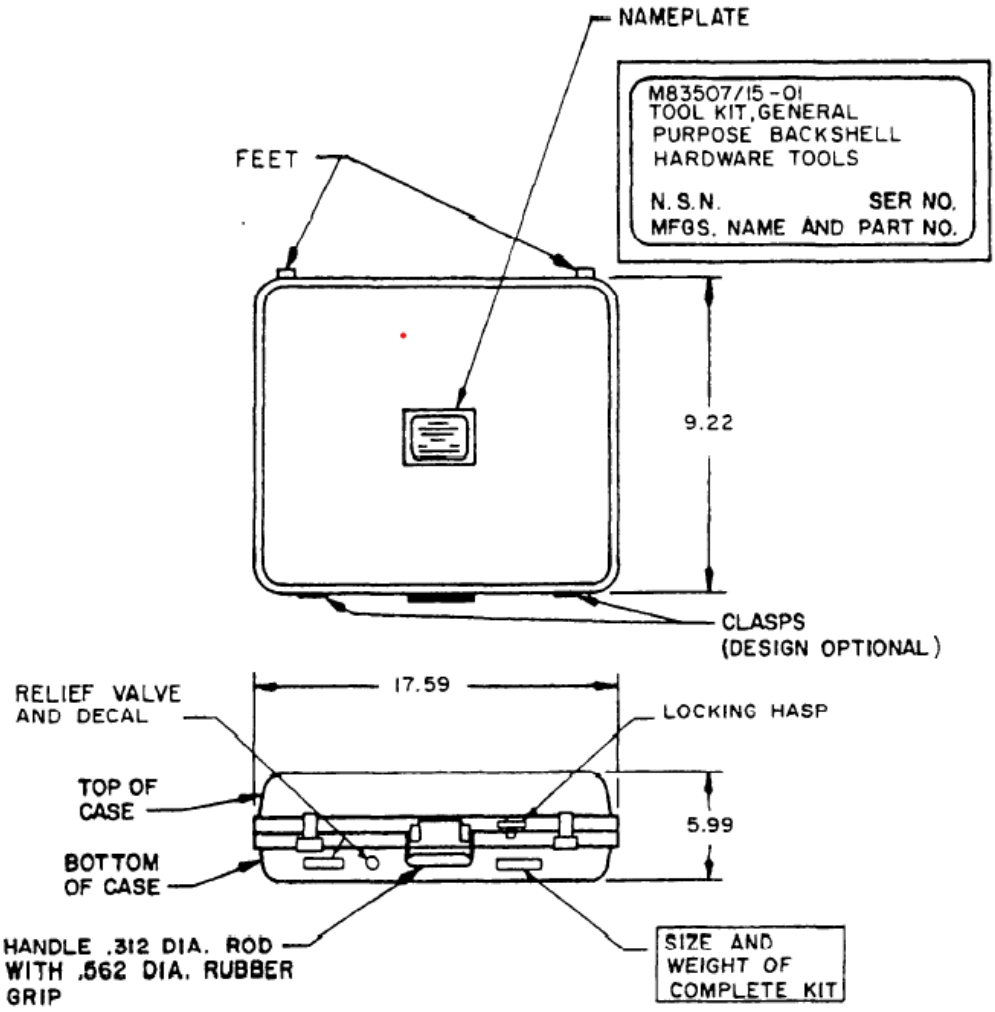


FIGURE 1 Metal case. Tool kit, general purpose, backshell hardware tools

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NOTES:

1. Part number M83507/15-01 includes case and tools.
2. Part number M83507/15-02 for case only (tools not included).

Part List:

(1)	M3507/15-02	Tool case assembly
(1)	Glenair G79005	Straight handle (1/4 drive each end) (or apporved equivalent)
(1)	Glenair G79006	Tee handle (1/4 drive) (or approved equivalent)
(1)	Glenair TG73	Strap wrench 1.25 (31.8 mm) maximum diameter (or approved equivalent)
(1)	Glenair TG74	Strap wrench 2.50 (63.5 mm) maximum diameter (or approved equivalent)
(1)	Glenair G79008	Adapter 3/8 to 1/4 drive (or approved equivalent)
(1)	Glenair G79007	Shaft extension 14.00 (355.6 mm) (or approved equivalent)
(1)	Glenair TG70	Strap wrench 3.00 inch (76.20 mm) maximum diameter (or approved equivalent)
(1)	Glenair TG69-P	Pliers, soft jaw (or approved equivalent)
(1)	Glenair G79010	Pencil, marking (or approved equivalent)
(1)	Glenair TG76-1	Pliers, parallel jaw (or approved equivalent)
(1)	Glenair TG76-2	Pliers, parallel jaw (or approved equivalent)
(1)	Glenair TG76-3	Pliers, parallel jaw (or approved equivalent)
(1)	Glenair TG76-4	Pliers, parallel jaw (or approved equivalent)
(1)	Glenair TG76-5	Pliers, parallel jaw (or approved equivalent)
(1)	Glenair TG76-6	Pliers, parallel jaw (or approved equivalent)
(1)	Glenair GTSQ150	Wrench torque (0 to 150 inch-pounds) (or approved equivalent)
(1)	SAE-AS90387	(R) Wiring installation tools for plastic and metal tiedown straps.
(1 set)		Contents charts, instruction sheets, and tool selection chart.

FIGURE 1 Metal case. Tool kit, general purpose, backshell hardware tools – Continued

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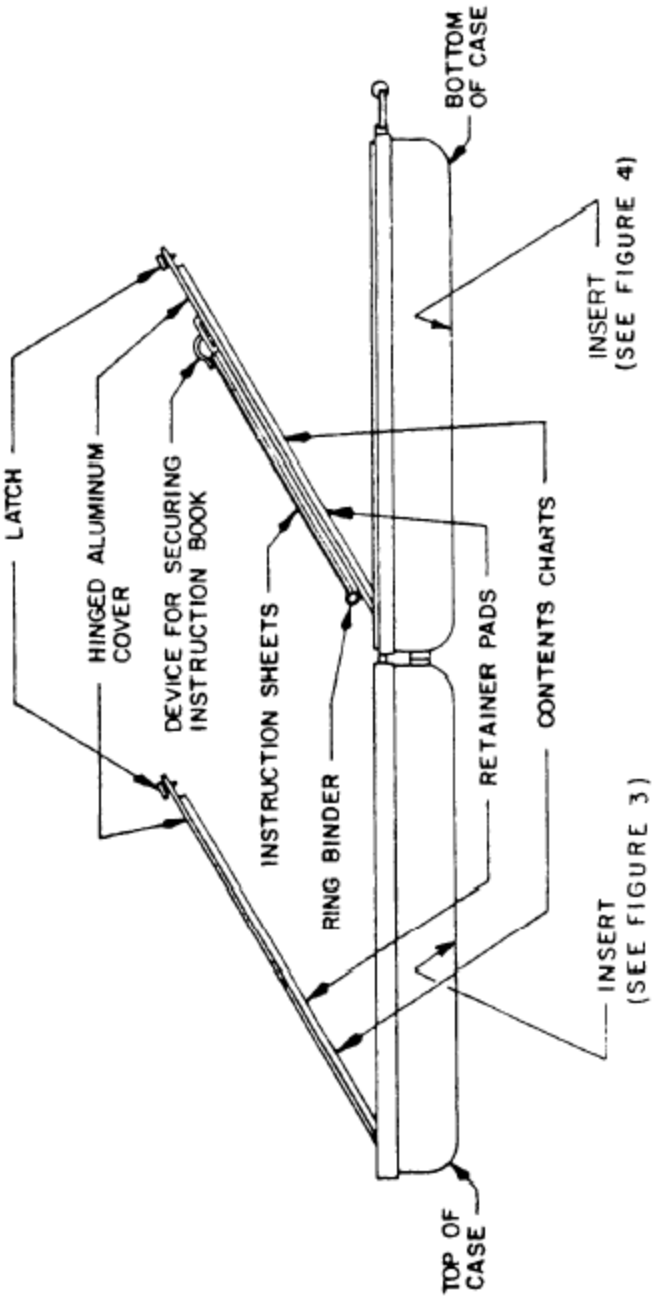


FIGURE 2 Metal case. Tool case assembly.

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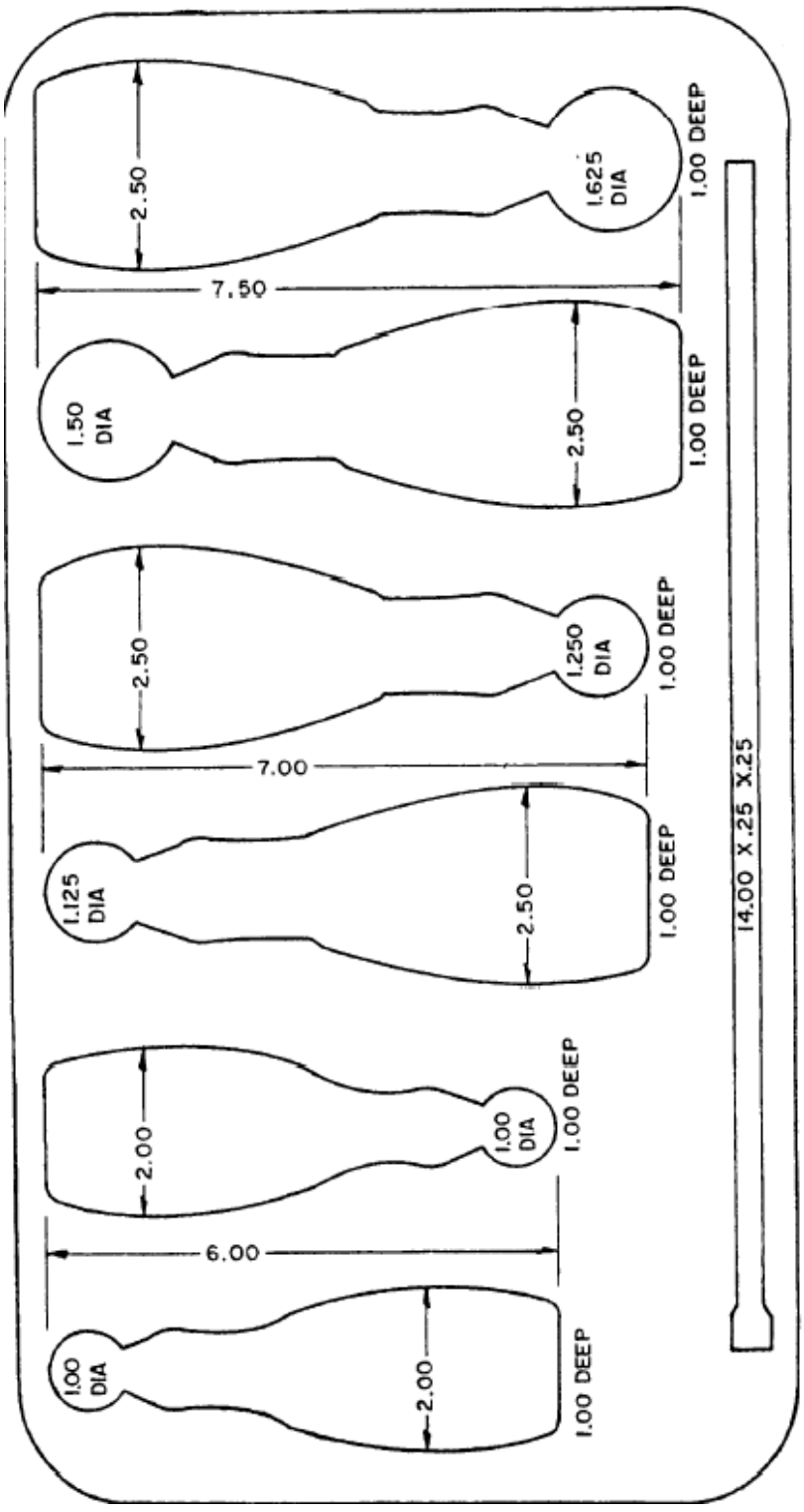


FIGURE 3 Metal case. Insert (top of case).

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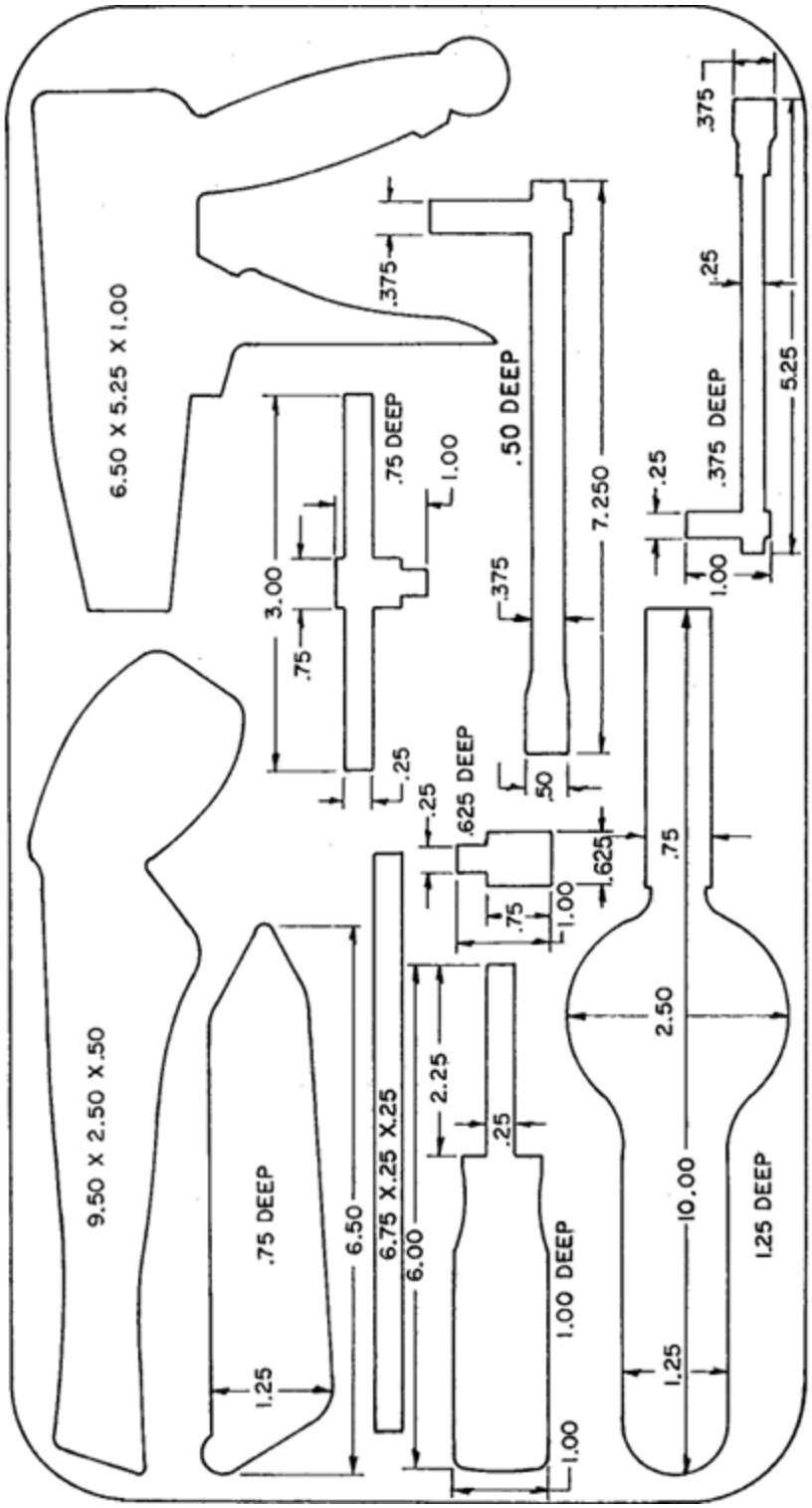


FIGURE 4 Metal case. Insert (bottom of case).

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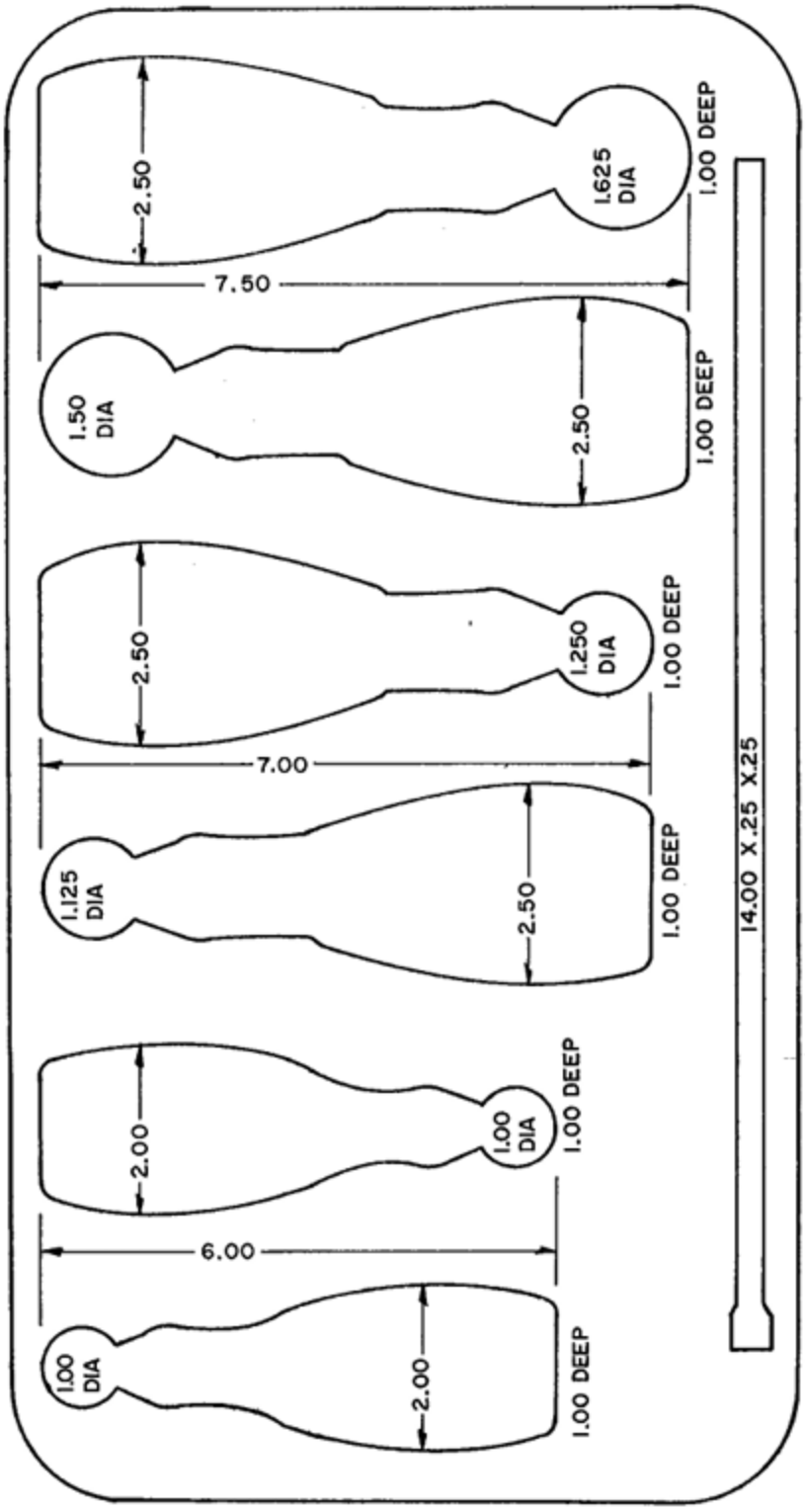


FIGURE 5 Metal case. Contents chart (top of case).

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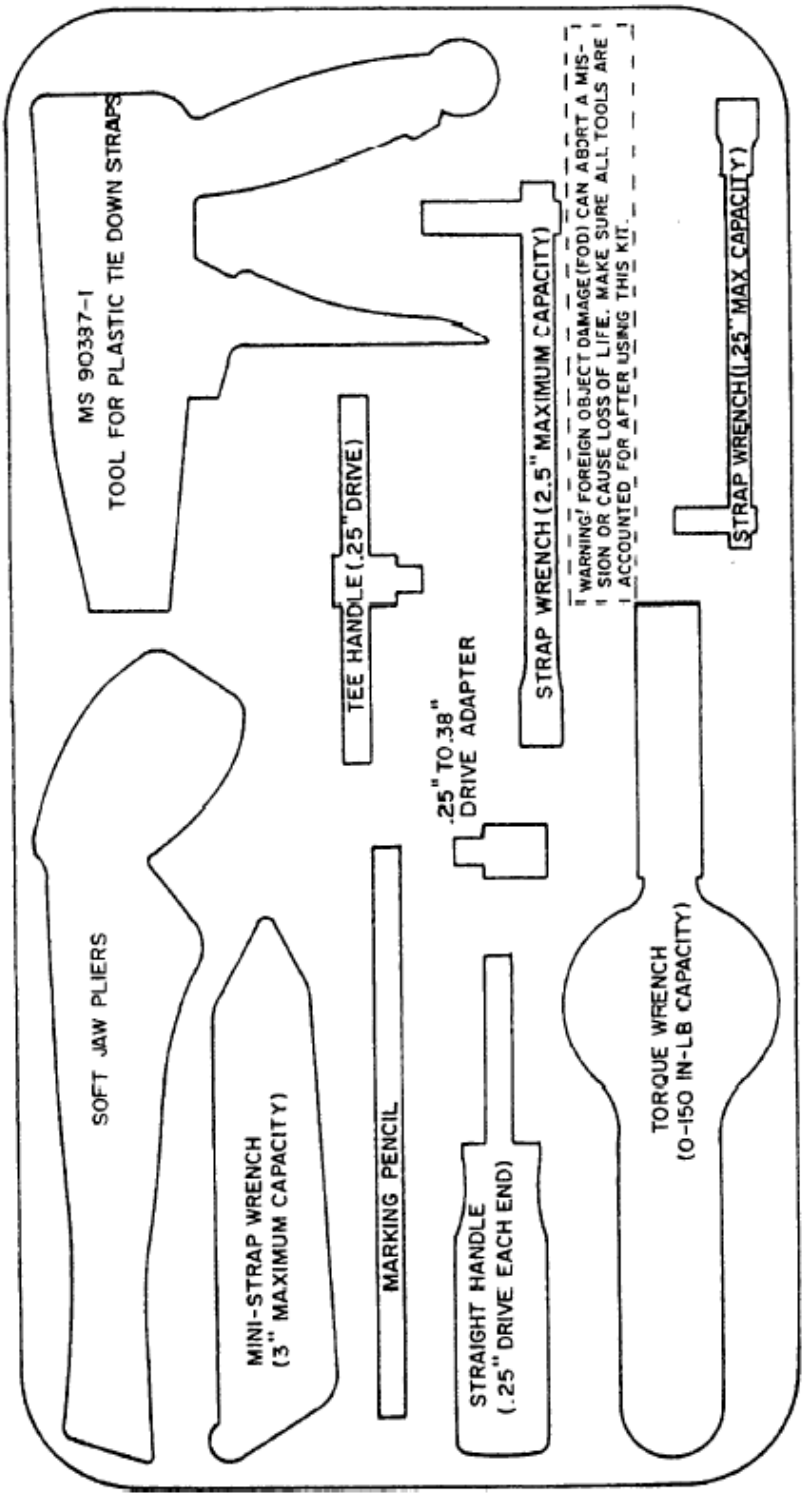


FIGURE 6 Metal case. Contents (bottom of case).

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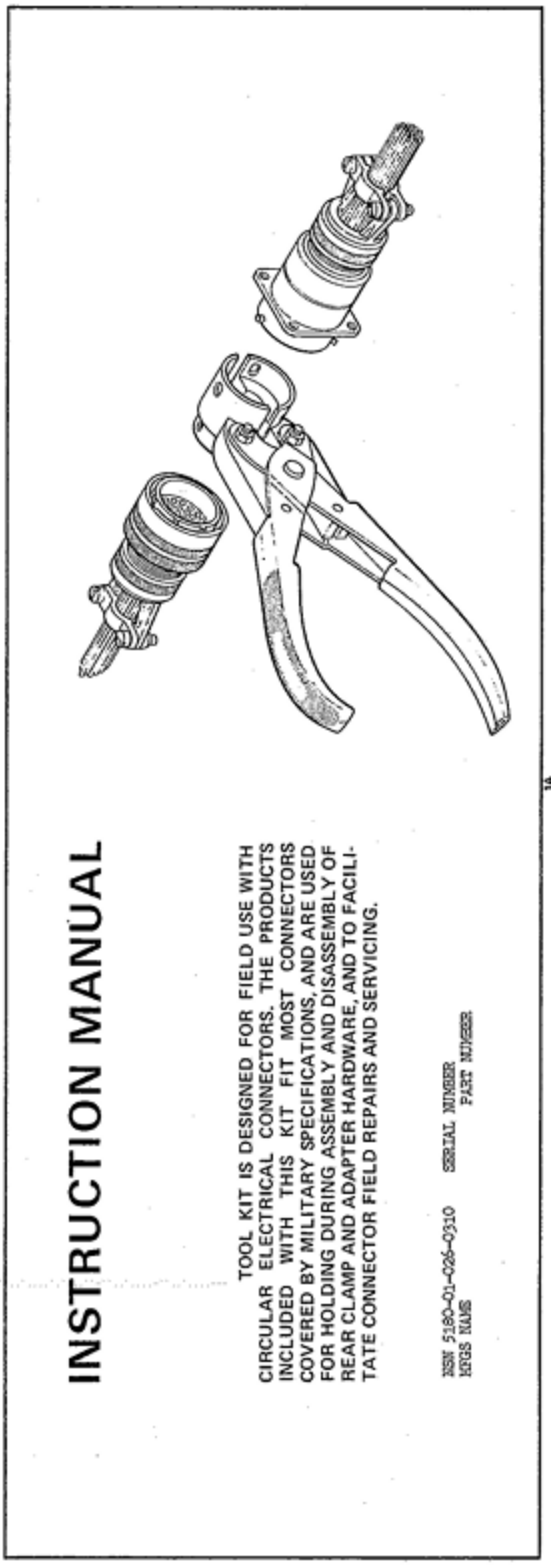


FIGURE 7 Metal case. Instruction sheet.

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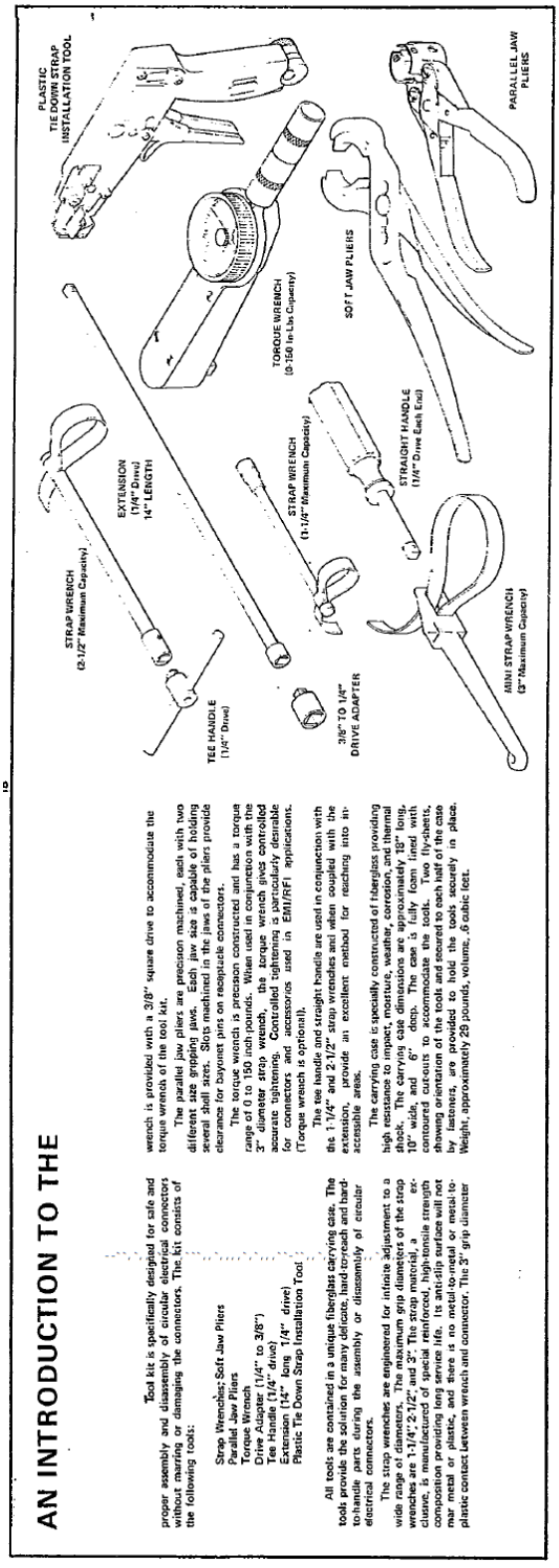


FIGURE 8 Metal case. Instruction sheet.

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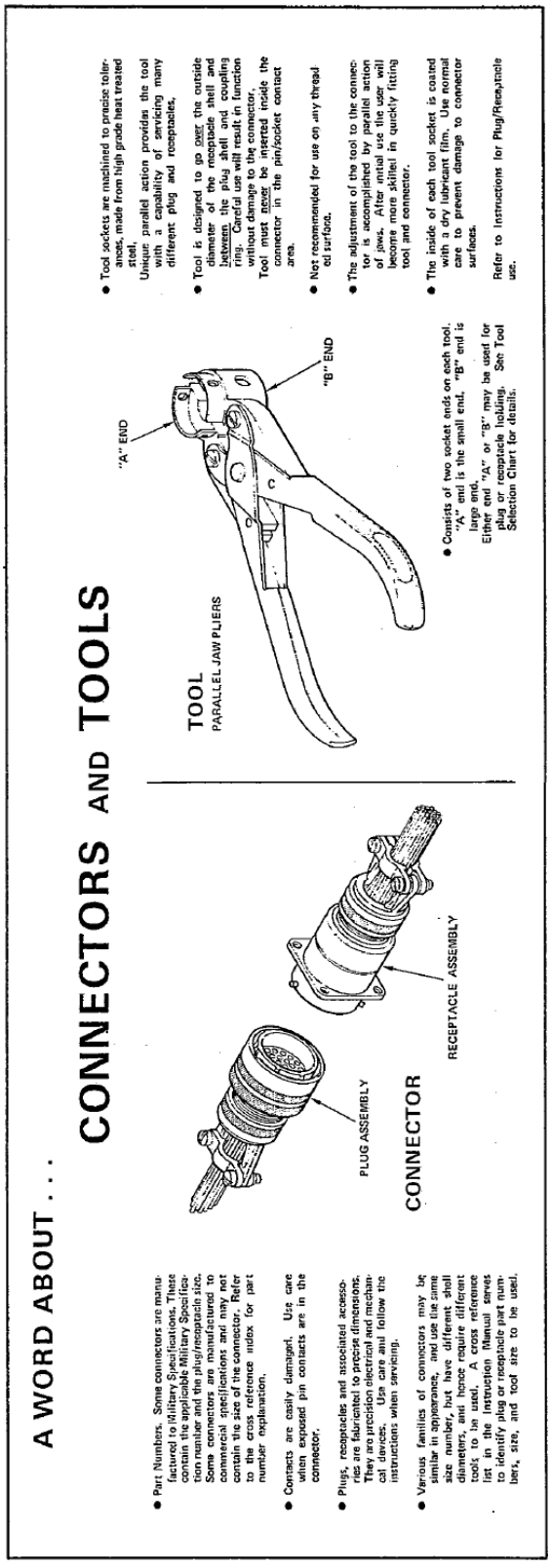


FIGURE 9 Metal case. Instruction sheet.

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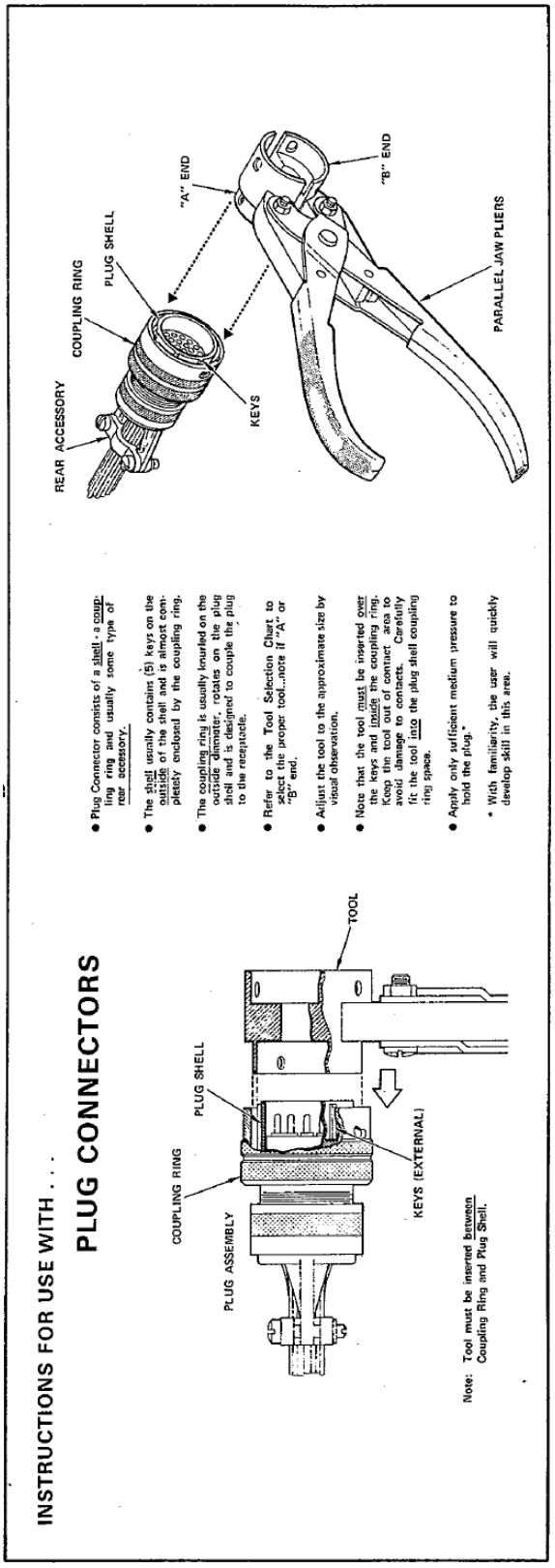


FIGURE 10 Metal case. Instruction sheet.

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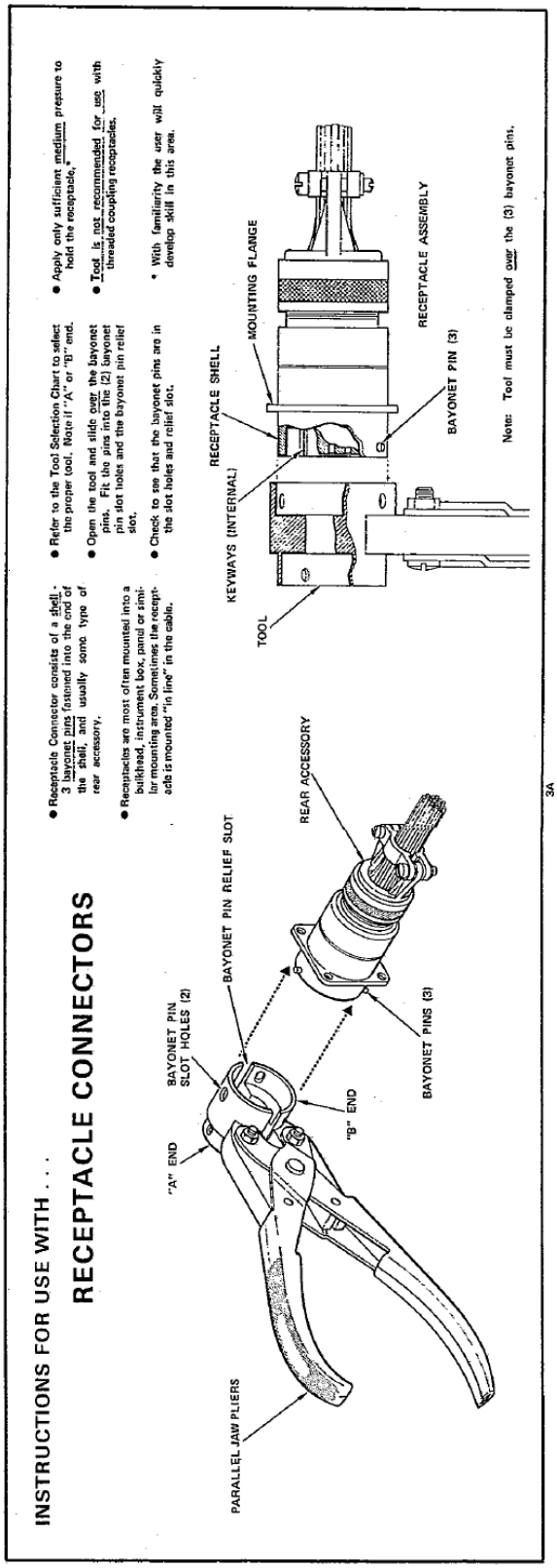


FIGURE 11 Metal case. Instruction sheet.

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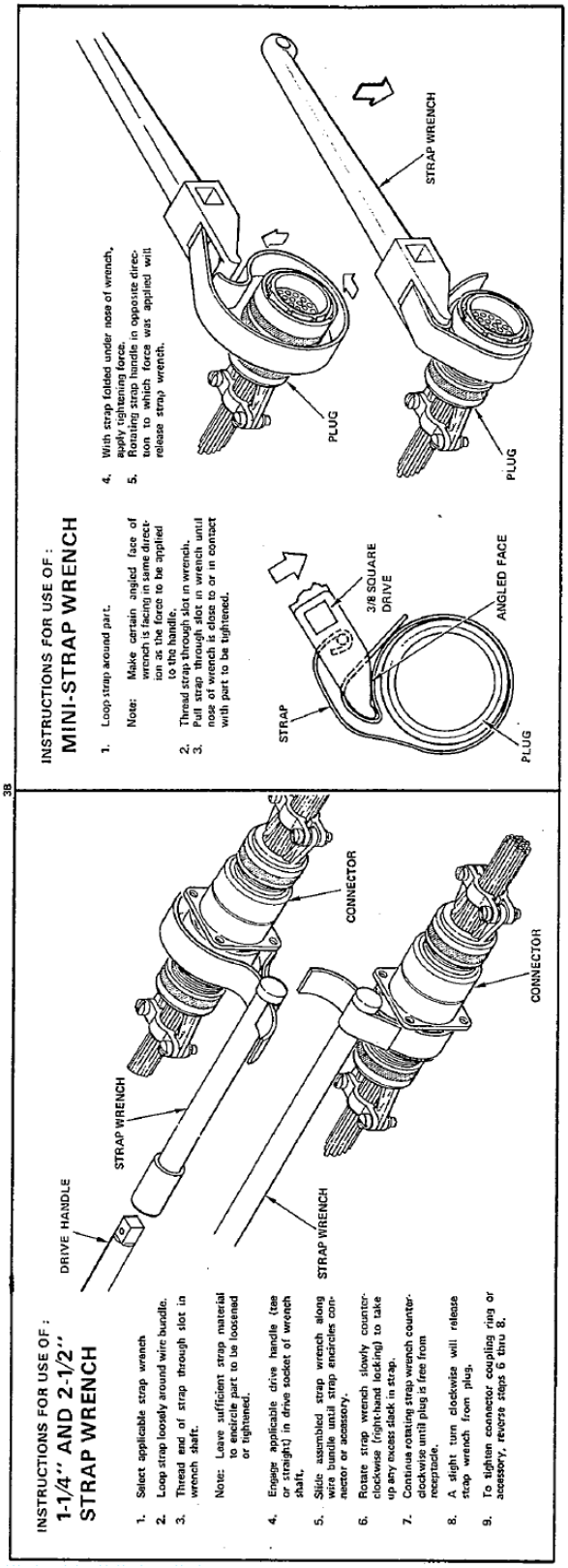


FIGURE 12 Metal case. Instruction sheet.

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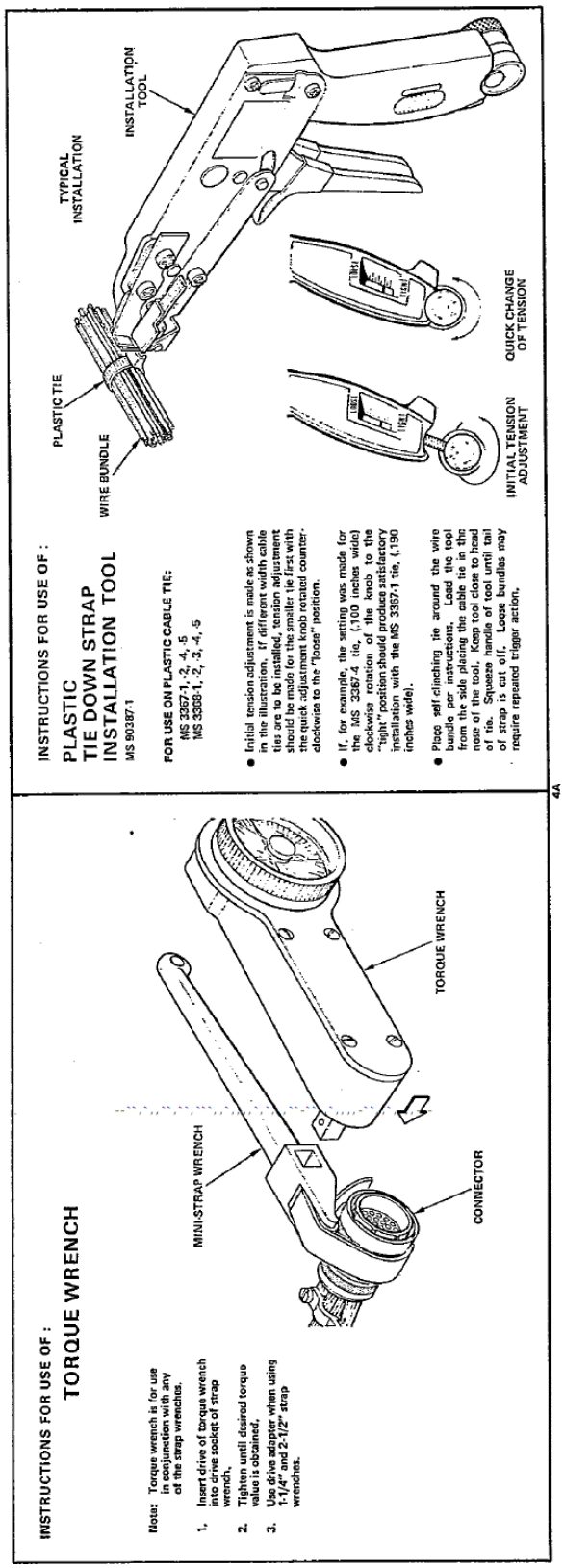
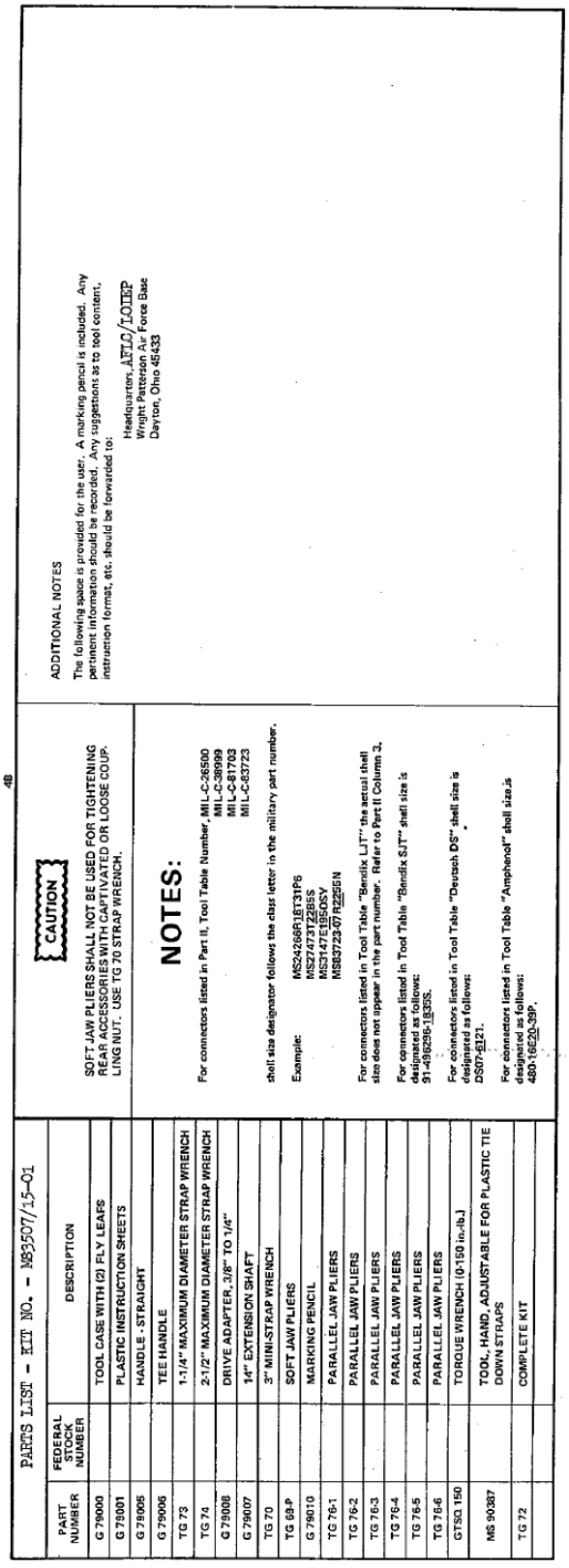


FIGURE 13 Metal case. Instruction sheet.

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PARTS LIST - KIT NO. - MS3507/15-01

PART NUMBER	FEDERAL STOCK NUMBER	DESCRIPTION
G 79000		TOOL CASE WITH (2) FLY LEAFS
G 79001		PLASTIC INSTRUCTION SHEETS
G 79005		HANDLE - STRAIGHT
G 79006		TEE HANDLE
TG 73		1-1/4" MAXIMUM DIAMETER STRAP WRENCH
TG 74		2-1/2" MAXIMUM DIAMETER STRAP WRENCH
G 79008		DRIVE ADAPTER, 3/8" TO 1/4"
G 79007		14" EXTENSION SHAFT
TG 70		3" MINI-STRAP WRENCH
TG 69-P		SOFT JAW PLIERS
G 79010		MARKING PENCIL
TG 76-1		PARALLEL JAW PLIERS
TG 76-2		PARALLEL JAW PLIERS
TG 76-3		PARALLEL JAW PLIERS
TG 76-4		PARALLEL JAW PLIERS
TG 76-5		PARALLEL JAW PLIERS
TG 76-6		PARALLEL JAW PLIERS
GTSQ 150		TORQUE WRENCH (0-150 in.-lb.)
MS 90387		TOOL, HAND, ADJUSTABLE FOR PLASTIC TIE DOWN STRAPS
TG 72		COMPLETE KIT

CAUTION

SOFT JAW PLIERS SHALL NOT BE USED FOR TIGHTENING REAR ACCESSORIES WITH CAPTIVATED OR LOOSE COUP-LING NUT. USE TG 70 STRAP WRENCH.

NOTES:

For connectors listed in Part II, Tool Table Number, MIL-C-26500 MIL-C-38999 MIL-C-81703 MIL-C-83723

shell size designator follows the class letter in the military part number.

Example: MS24266R18T31P6
MS27473T22B8S
MS3147E16S0SY
M83723-07R2256N

For connectors listed in Tool Table "Bendix LJT" the actual shell size does not appear in the part number. Refer to Part II Column 3.

For connectors listed in Tool Table "Bendix SJT" shell size is designated as follows: 91-496256-18S8S.

For connectors listed in Tool Table "Deutsch DS" shell size is designated as follows: DS07-6121.

For connectors listed in Tool Table "Amphenol" shell size is designated as follows: 480-16E20-39P.

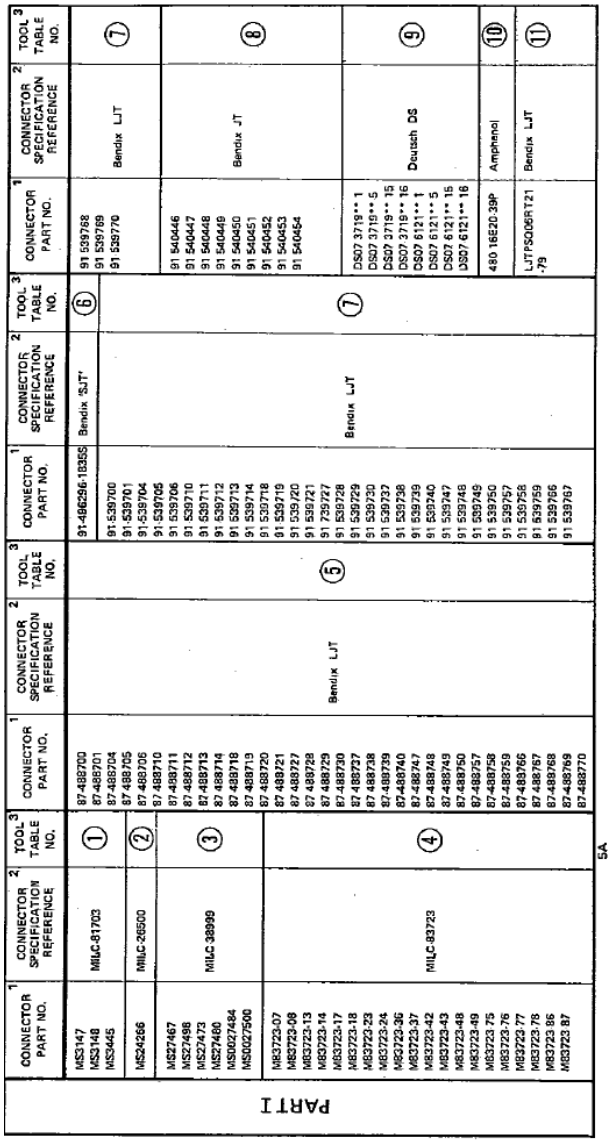
ADDITIONAL NOTES

The following space is provided for the user. A marking pencil is included. Any pertinent information should be recorded. Any suggestions as to tool content, instruction format, etc. should be forwarded to:

Headquarters, AFLC/LOEP
Wright Patterson Air Force Base
Dayton, Ohio 45433

FIGURE 14 Metal case. Instruction sheet.

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PART I

CONNECTOR PART NO. [1]	CONNECTOR SPECIFICATION REFERENCE [2]	TOOL TABLE NO. [3]	CONNECTOR PART NO. [1]	CONNECTOR SPECIFICATION REFERENCE [2]	TOOL TABLE NO. [3]	CONNECTOR PART NO. [1]	CONNECTOR SPECIFICATION REFERENCE [2]	TOOL TABLE NO. [3]	CONNECTOR PART NO. [1]	CONNECTOR SPECIFICATION REFERENCE [2]	TOOL TABLE NO. [3]
MS3147	MilC-81703	①	87-488700			91-486256-18355	Bendix "SJT"	⑥	91 539768		
MS3148			87-488701						91 539769		
MS3445			87-488704			91 539700			91 539770	Bendix LJT	⑦
			87-488705			91-539701					
MS24266	MilC-26500	②	87-488706			91-539704					
			87-488710			91 539705					
MS27467			87-488711			91 539706			91 540446		
MS27498			87-488712			91 539710			91 540447		
MS27473	MilC 38999	③	87-488713			91 539711			91 540448		
MS27480			87-488714			91 539712			91 540449	Bendix JT	⑧
MS90027484			87-488718			91 539713			91 540450		
MS90027500			87-488719			91 539714			91 540451		
			87-488720			91 539718			91 540452		
M83723-07			87-488721			91 539719			91 540453		
M83723-08			87-488727			91 539720			91 540454		
M83723-13			87-488728			91 539721					
M83723-14			87-488729			91 739727					
M83723-17			87-488730			91 539728			DS07 3719** 1		
M83723-18			87-488737			91 539729			DS07 3719** 5		
M83723-23			87-488738			91 539730			DS07 3719** 15		
M83723-24			87-488739			91 539733			DS07 3719** 16		
M83723-36			87-488740			91 539738			DS07 6121** 1	Deutsch DS	⑨
M83723-37			87-488747			91 539739			DS07 6121** 5		
M83723-42	MilC 83723	④	87-488748			91 539740			DS07 6121** 15		
M83723-43			87-488749			91 539747			DS07 6121** 16		
M83723-48			87-488750			91 539748					
M83723-49			87-488757			91 539749					
M83723-75			87-488758			91 539750			480 16E20 39P	Amphenol	⑩
M83723-76			87-488759			91 539757					
M83723-77			87-488766			91 539758			LJT FSQ06RT21 -79	Bendix LJT	⑪
M83723-78			87-488767			91 539759					
M83723 86			87-488768			91 539766					
M83723 87			87-488770			91 539767					

NOTES ON TOOL SELECTION CHART

The following pages provide a crossreference between connectors and tools required to service them. The chart is intended as a guide, and some special connectors may not be covered.

The Tool Selection Chart is divided into two parts. Part I provides cross-reference between connector part numbers and tool table numbers; Part II provides a cross-reference between Tool Table Number, Connector Part Number and Shell Size and Tool Size.

Part I For selection of the correct tool table number proceed as follows:

1. Determine connector part number.
2. In column 1, find the applicable connector part number.
3. Column 2 references Military or Supplier Specification.
4. Reading across the columns, column 3 indicates the correct Tool Table Number to be used.
5. Turn to Part II, Tool Table Numbers.

Part II For selection of the correct tool size, proceed as follows:

1. Find the required Tool Table Number in Column 1.
2. In column 2, find the applicable connector part number (connector part numbers are listed in alpha-numeric sequence).
3. To determine shell size from connector part number, see notes for example.
4. Find shell size in column 3.
5. Column 4 indicates the tool size and A or B socket end to be used.

FIGURE 15 Metal case. Instruction sheet.

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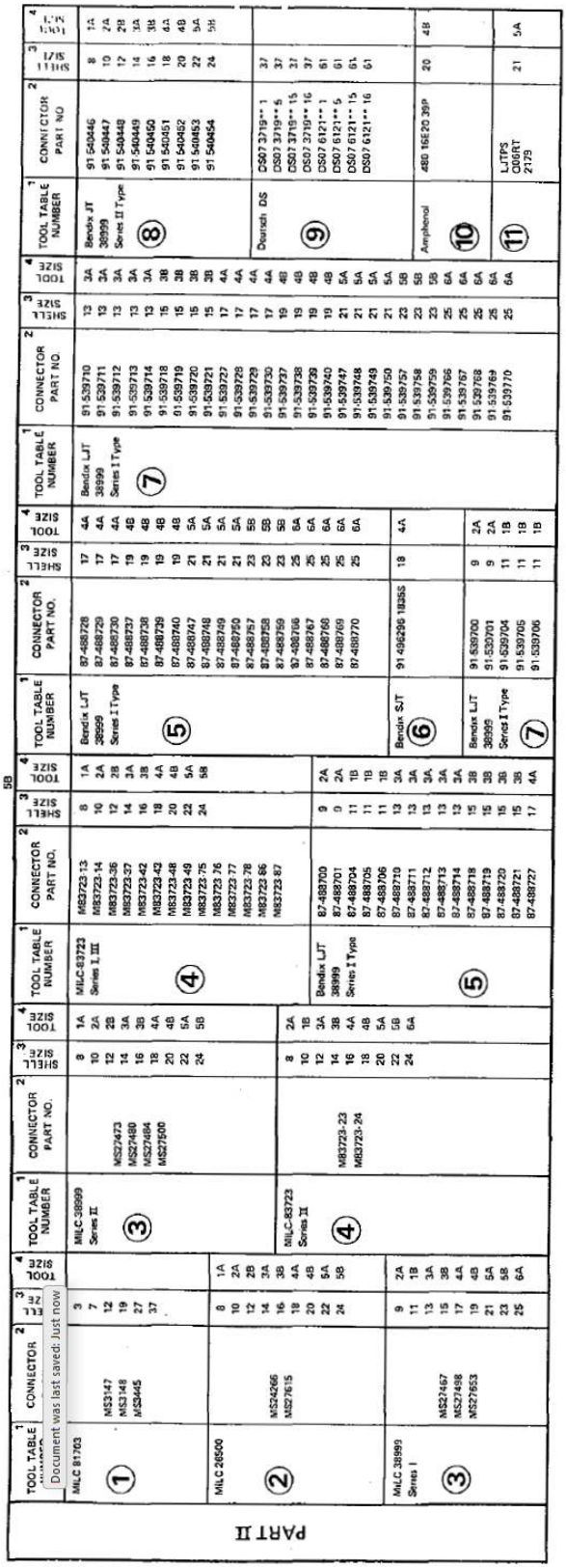


FIGURE 16 Metal case. Instruction sheet

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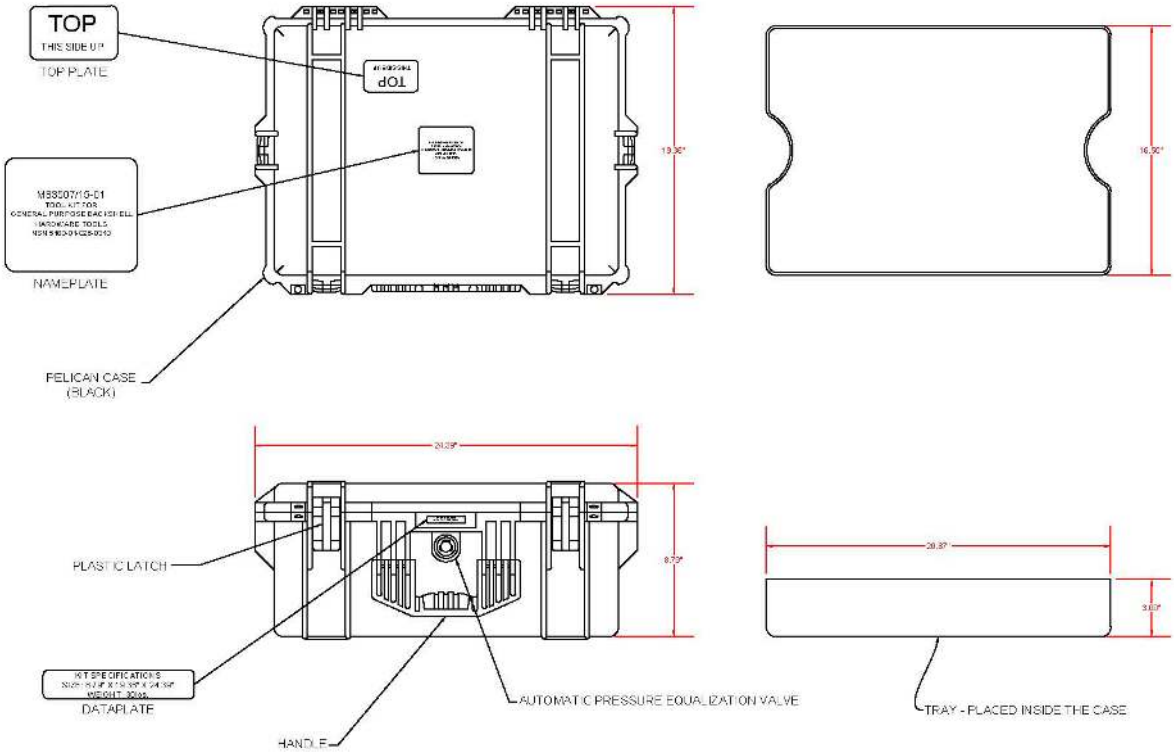


FIGURE 17 Plastic case. TOOL KIT FOR GENERAL PURPOSE BACKSHELL HARDWARE.

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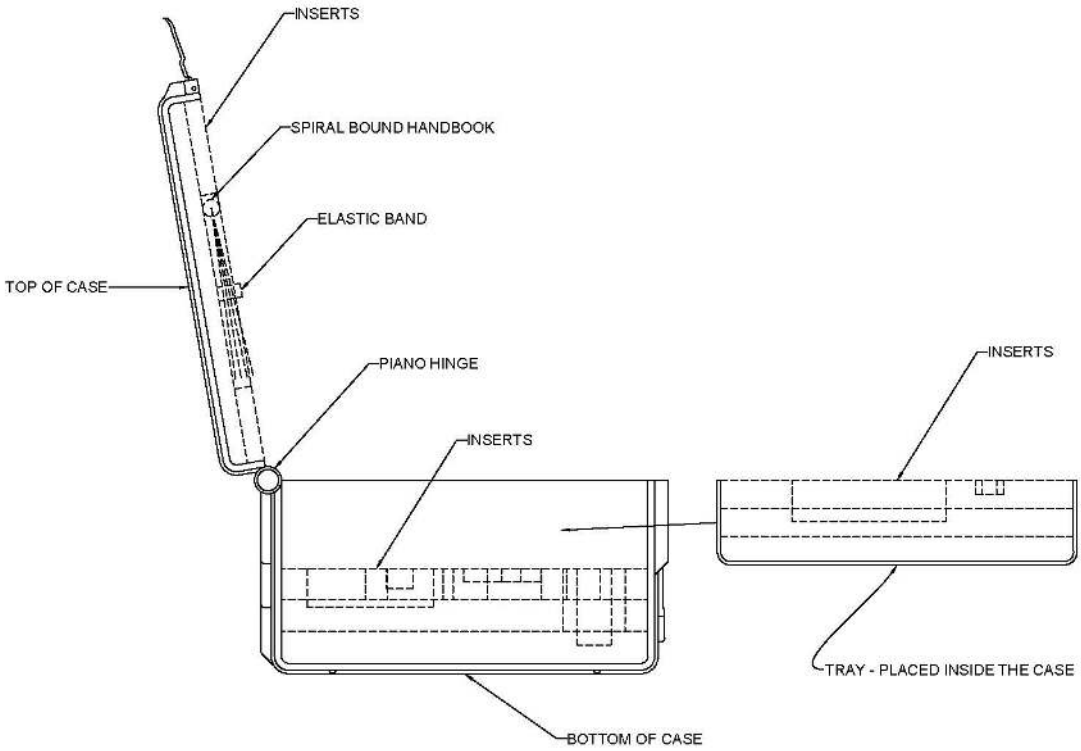


FIGURE 18 Plastic case. TOOL CASE ASSEMBLY.

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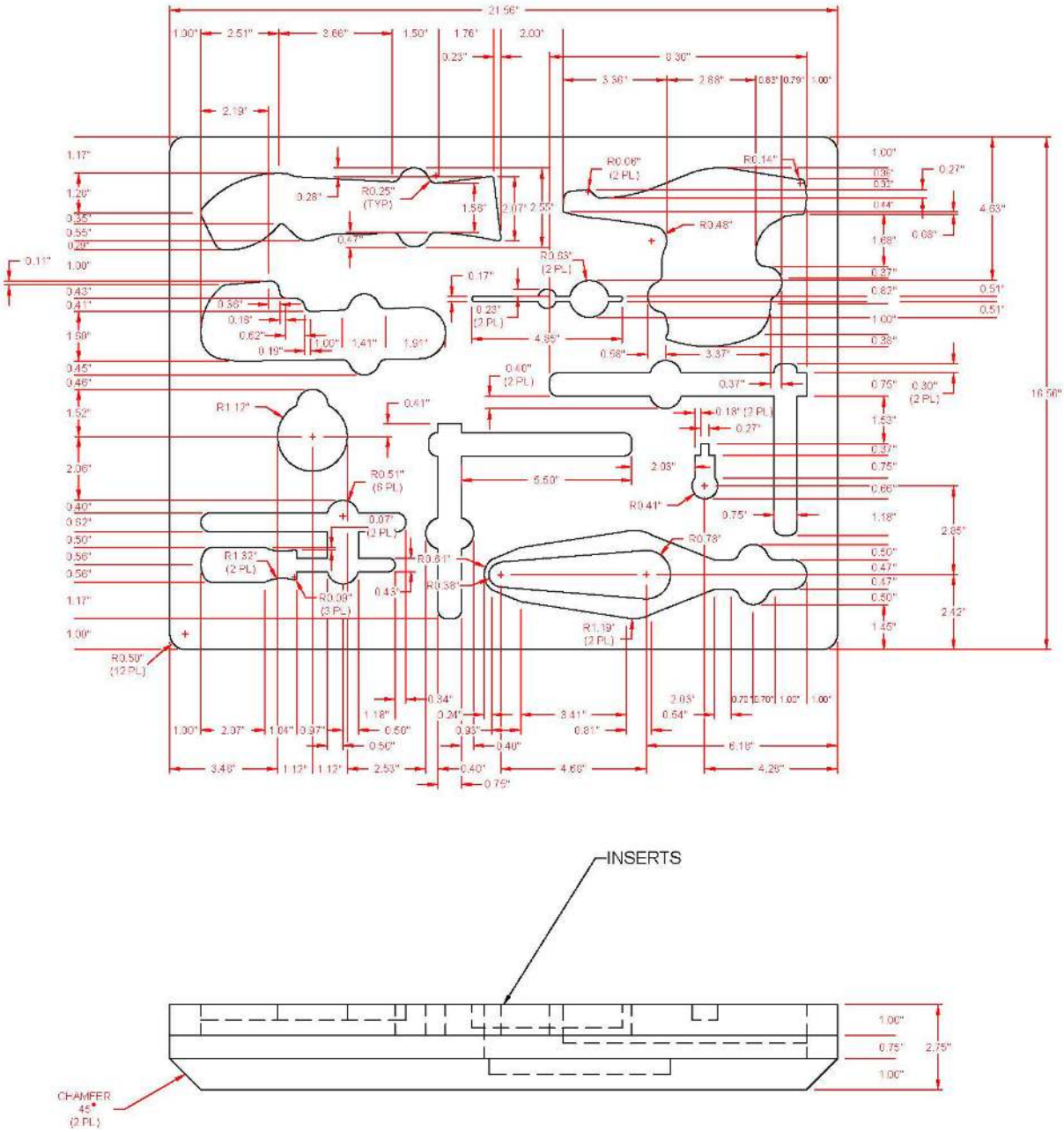


FIGURE 19 Plastic case. INSERT, BOTTOM OF CASE

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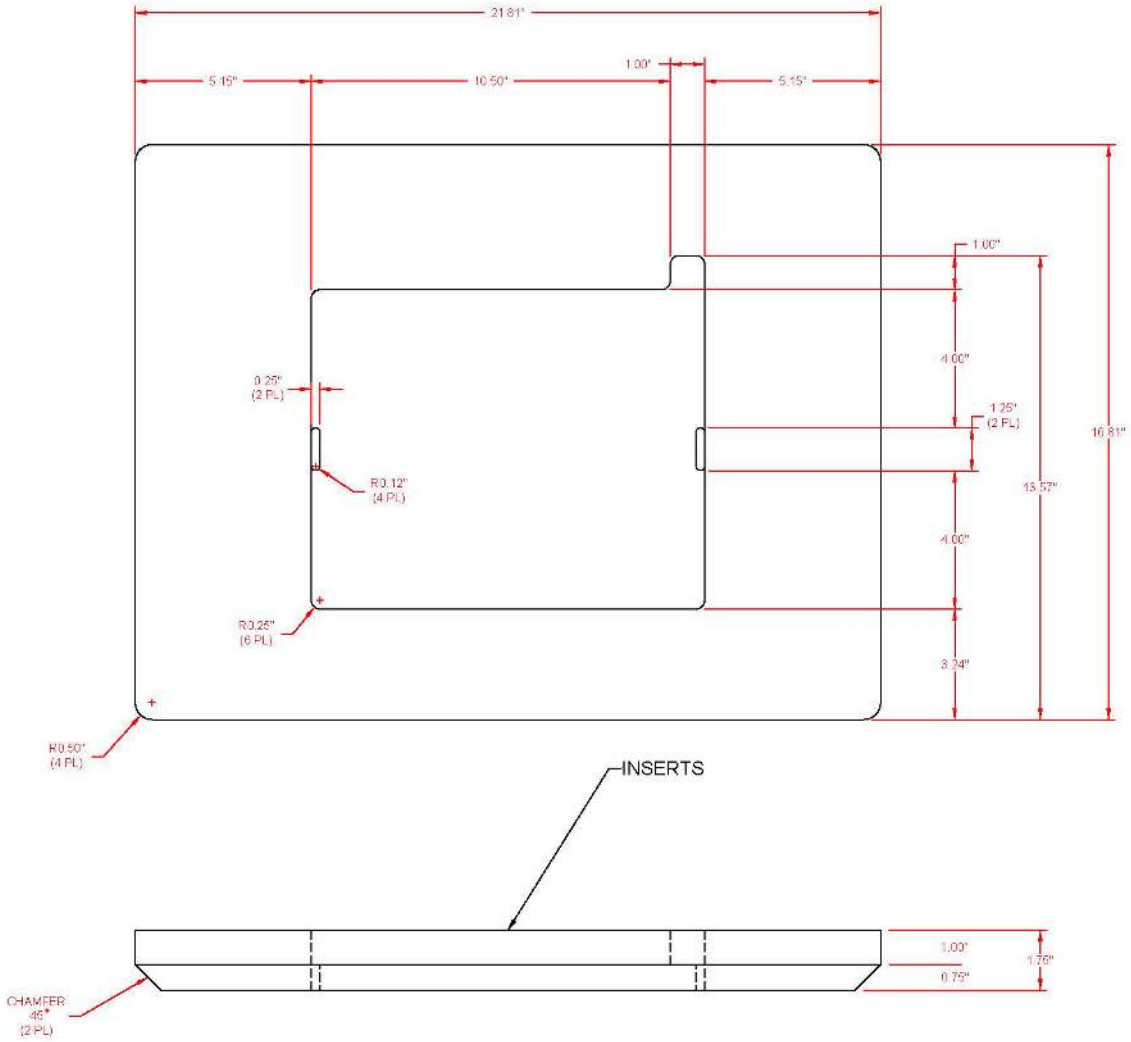


FIGURE 20 Plastic case. INSERT, TOP OF CASE

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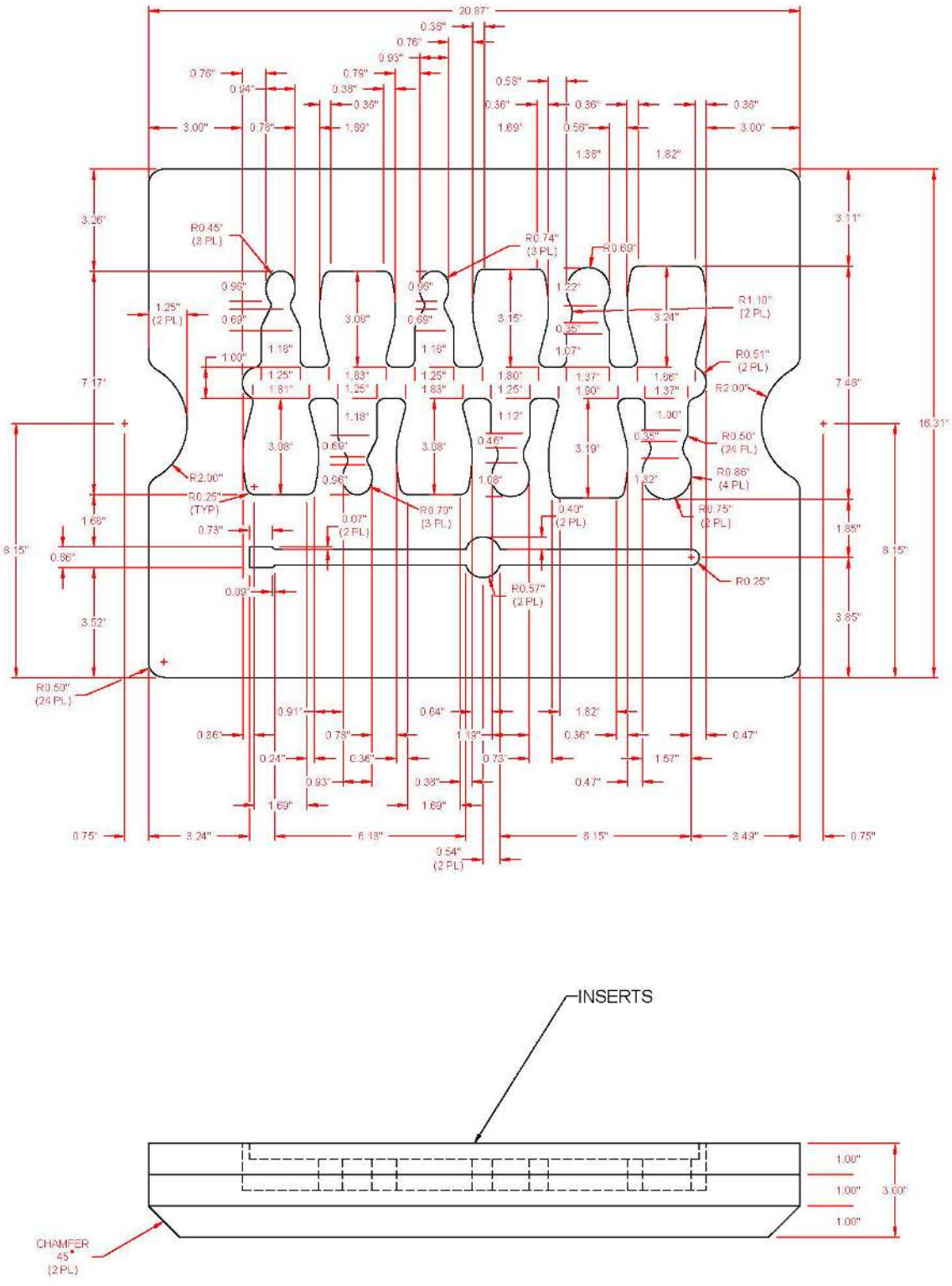


FIGURE 21 Plastic case. INSERT TRAY.

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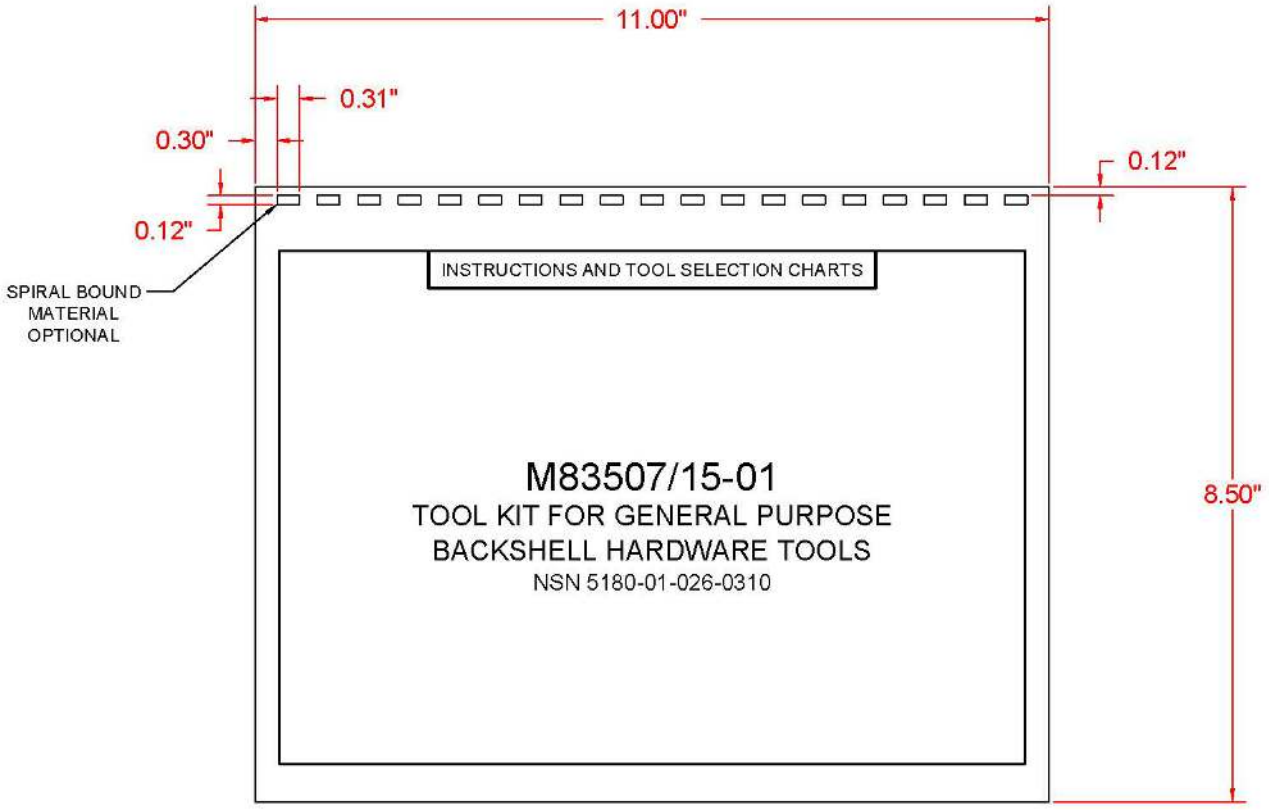


FIGURE 22 Plastic case. Instruction sheet.

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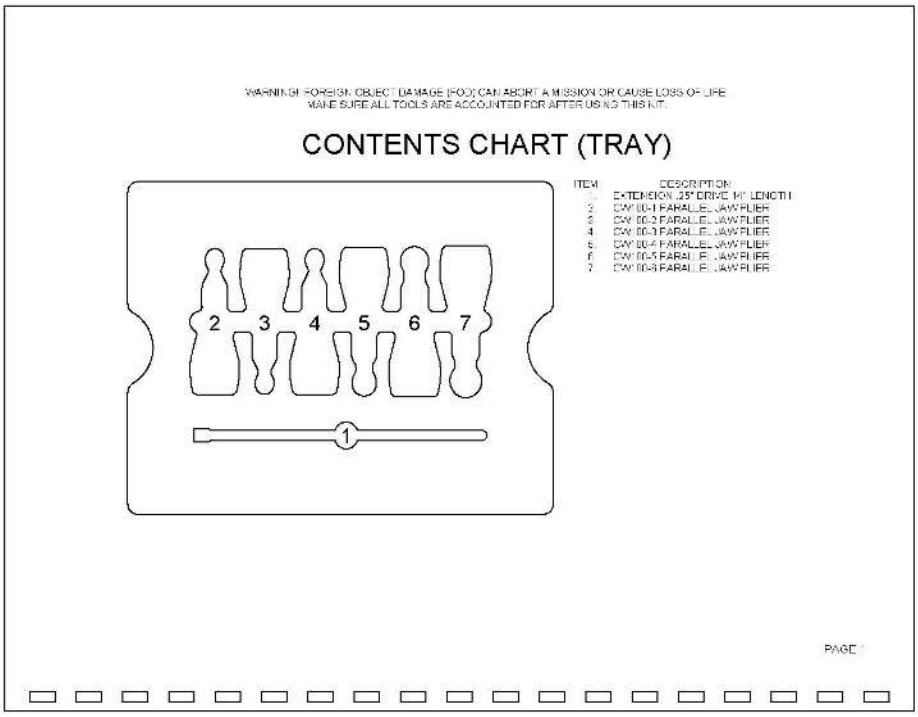


FIGURE 23 Plastic case. Instruction sheet.

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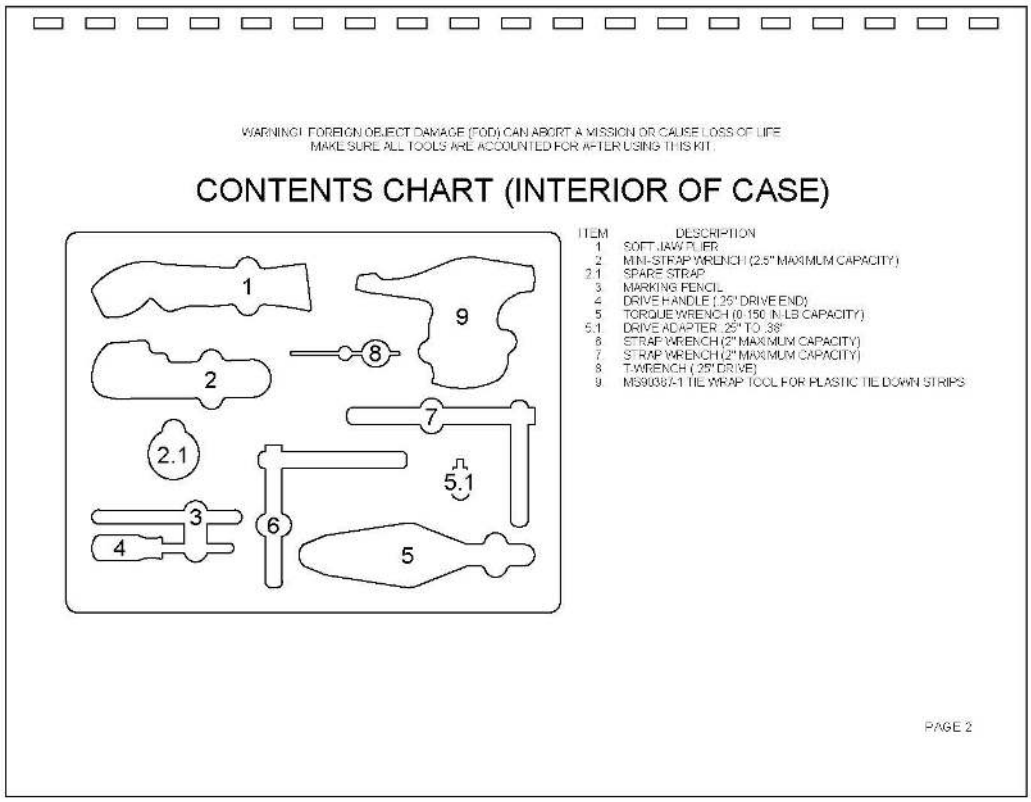


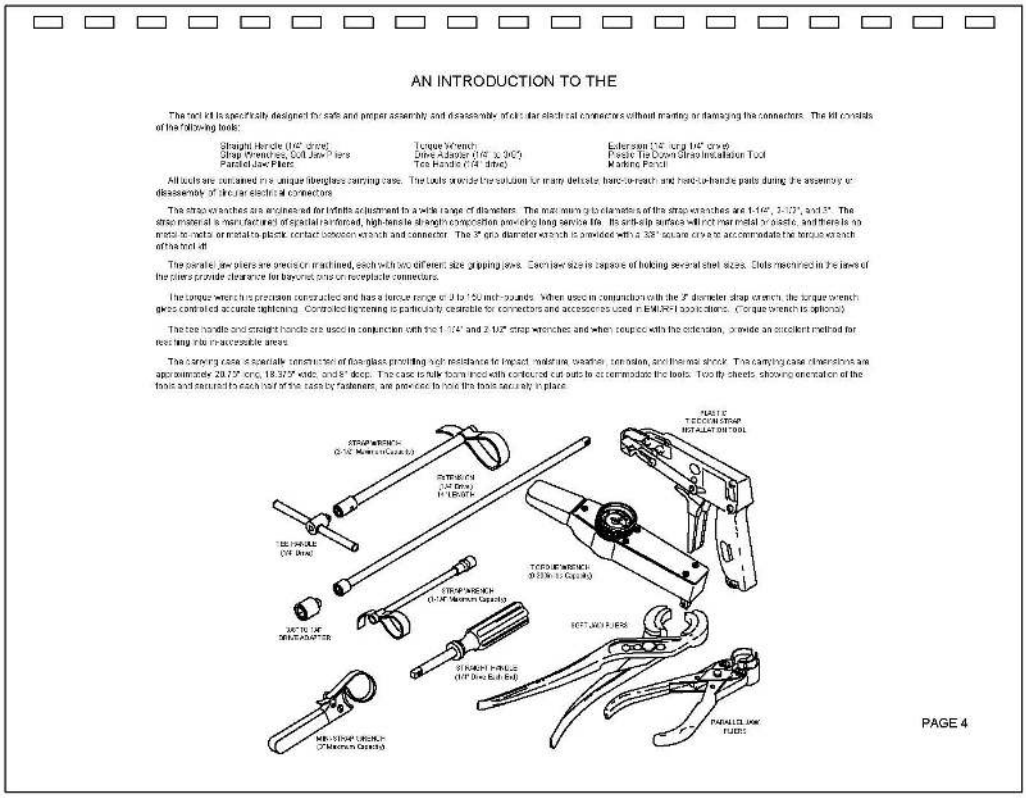
FIGURE 24 Plastic case. Instruction sheet.

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FIGURE 25 Plastic case. Instruction sheet.

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AN INTRODUCTION TO THE

The tool kit is specifically designed for safe and proper assembly and disassembly of circular electrical connectors without marring or damaging the connectors. The kit consists of the following tools:

Straight Handle (1/4" drive)	Torque Wrench	Extension (12" long 1/4" drive)
Strap Wrenches, Split Jaw Pliers	Drive Adapter (1/4" to 3/8")	Plastic Tie Down Strap Installation Tool
Parallel Jaw Pliers	Tee Handle (1/4" drive)	Marking Pencil

All tools are contained in a unique fiberglass carrying case. The tools provide the solution for many delicate, hard-to-reach and hard-to-handle parts during the assembly or disassembly of circular electrical connectors.

The strap wrenches are engineered for infinite adjustment to a wide range of diameters. The maximum grip diameters of the strap wrenches are 1-1/4", 2-1/2", and 3". The strap material is manufactured of spatial reinforced, high-tensile strength composition providing long service life. Its anti-slip surface will not mar metal or plastic, and there is no metal-to-metal or metal-to-plastic contact between wrench and connector. The 3" grip diameter wrench is provided with a 3/8" square drive to accommodate the torque wrench of the tool kit.

The parallel jaw pliers are precision machined, each with two different size gripping jaws. Each jaw size is capable of holding several shell sizes. Slots machined in the jaws of the pliers provide clearance for bayonet pins on receptacle connectors.

The torque wrench is precision constructed and has a torque range of 0 to 150 inch-pounds. When used in conjunction with the 3" diameter strap wrench, the torque wrench gives controlled accurate tightening. Controlled tightening is particularly desirable for connectors and accessories used in EMI/RFI applications. (Torque wrench is optional)

The tee handle and straight handle are used in conjunction with the 1-1/4" and 2-1/2" strap wrenches and when coupled with the extension, provide an excellent method for reaching into inaccessible areas.

The carrying case is specially constructed of fiberglass providing high resistance to impact, moisture, weather, corrosion, and thermal shock. The carrying case dimensions are approximately 20.75" long, 18.375" wide, and 8" deep. The case is fully foam lined with contoured cut outs to accommodate the tools. Tool by sheets, showing orientation of the tools and secured to each half of the case by fasteners, are provided to hold the tools securely in place.

PAGE 4

FIGURE 26 Plastic case. Instruction sheet.

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Check the source to verify that this is the current version before use.

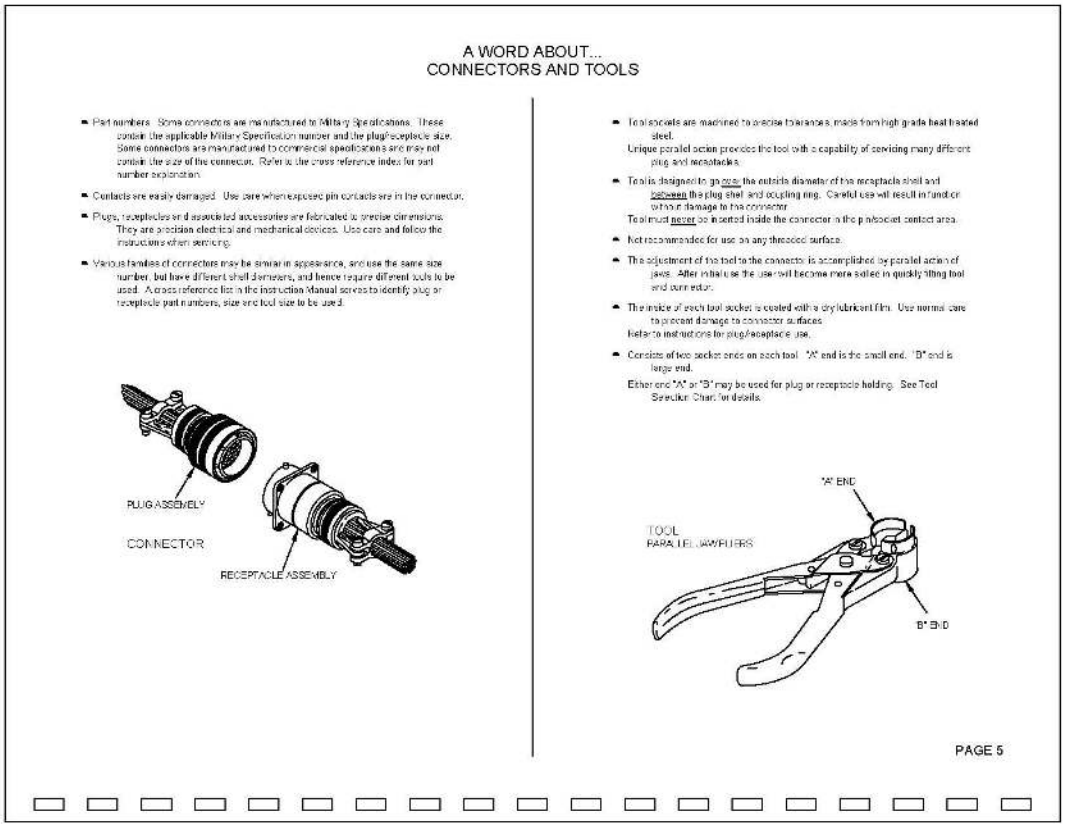


FIGURE 27 Plastic case. Instruction sheet.

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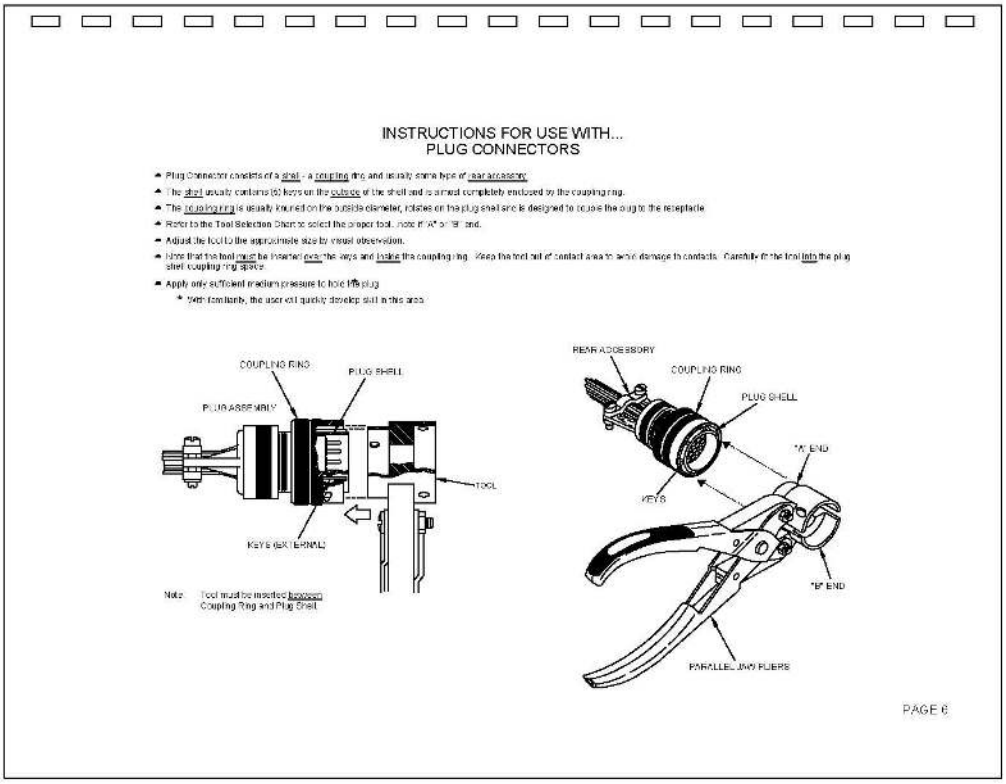


FIGURE 28 Plastic case. Instruction sheet.

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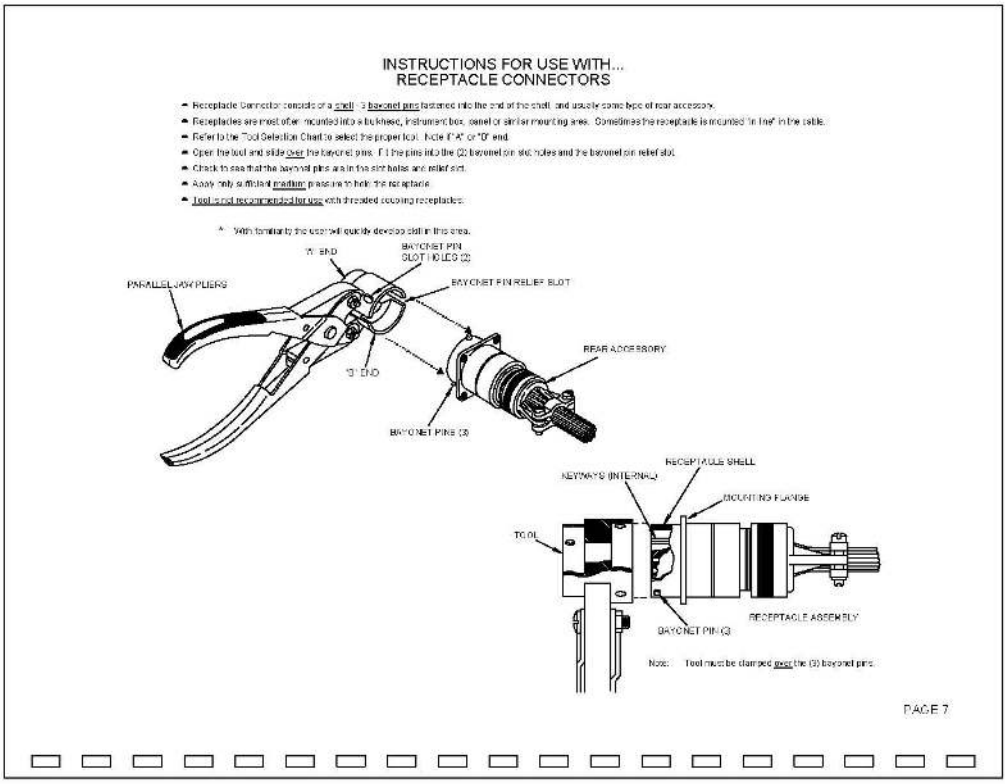


FIGURE 29 Plastic case. Instruction sheet.

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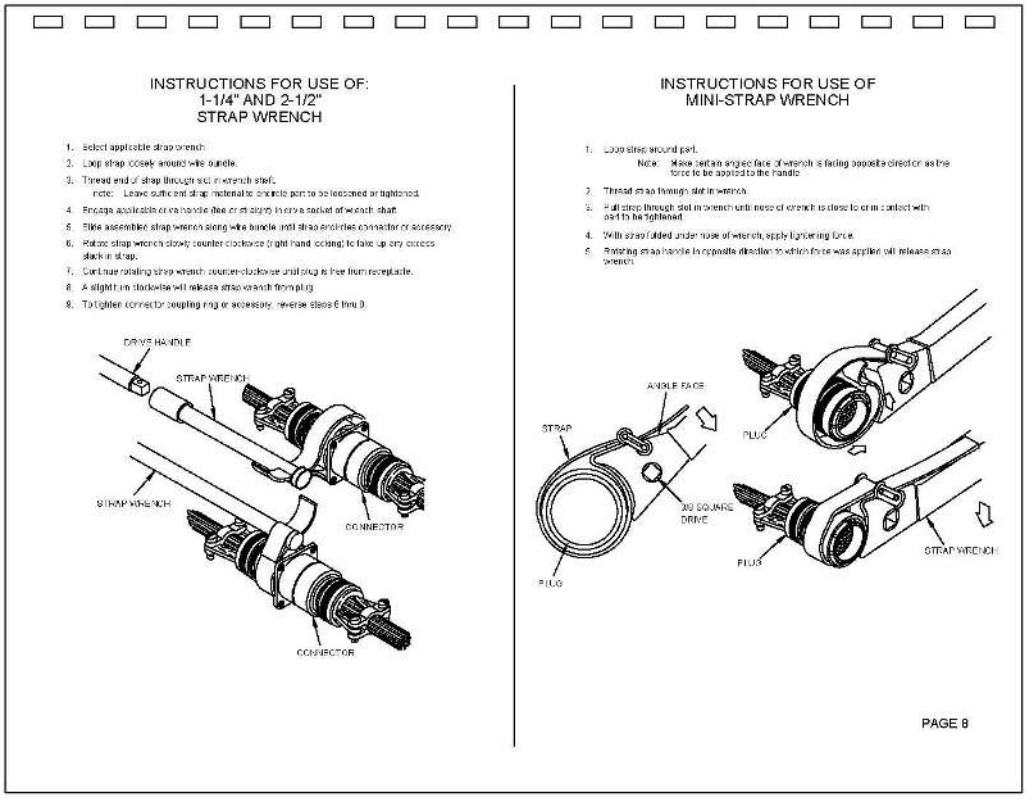


FIGURE 30 Plastic case. Instruction sheet.

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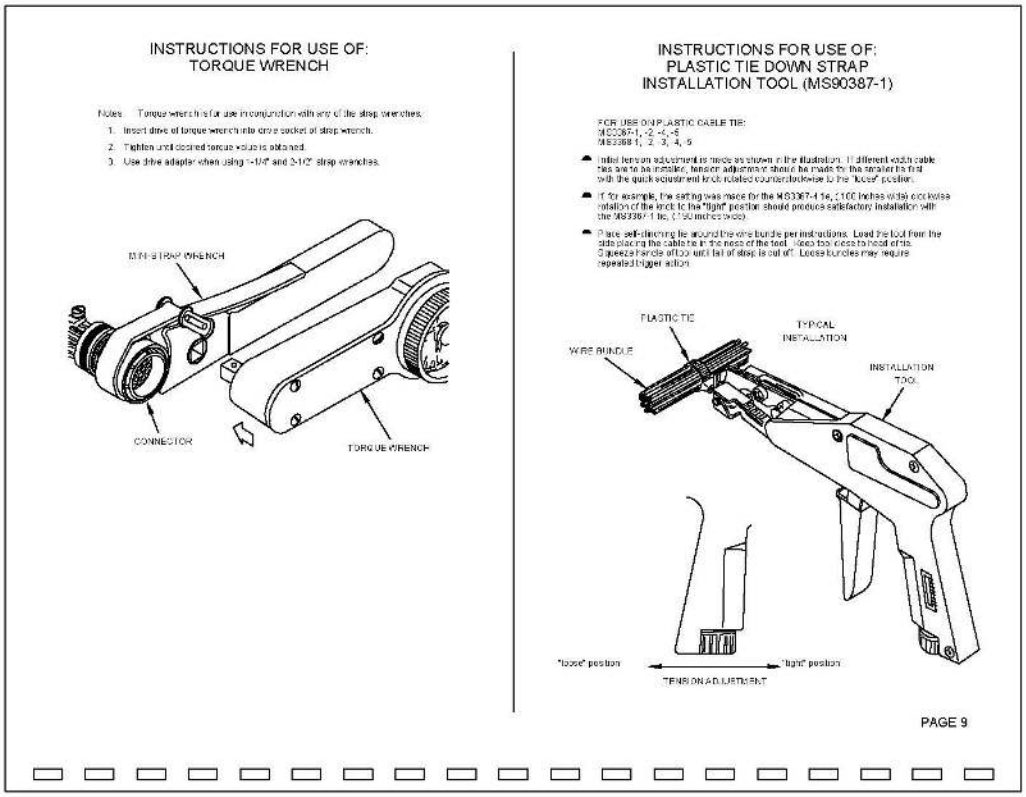
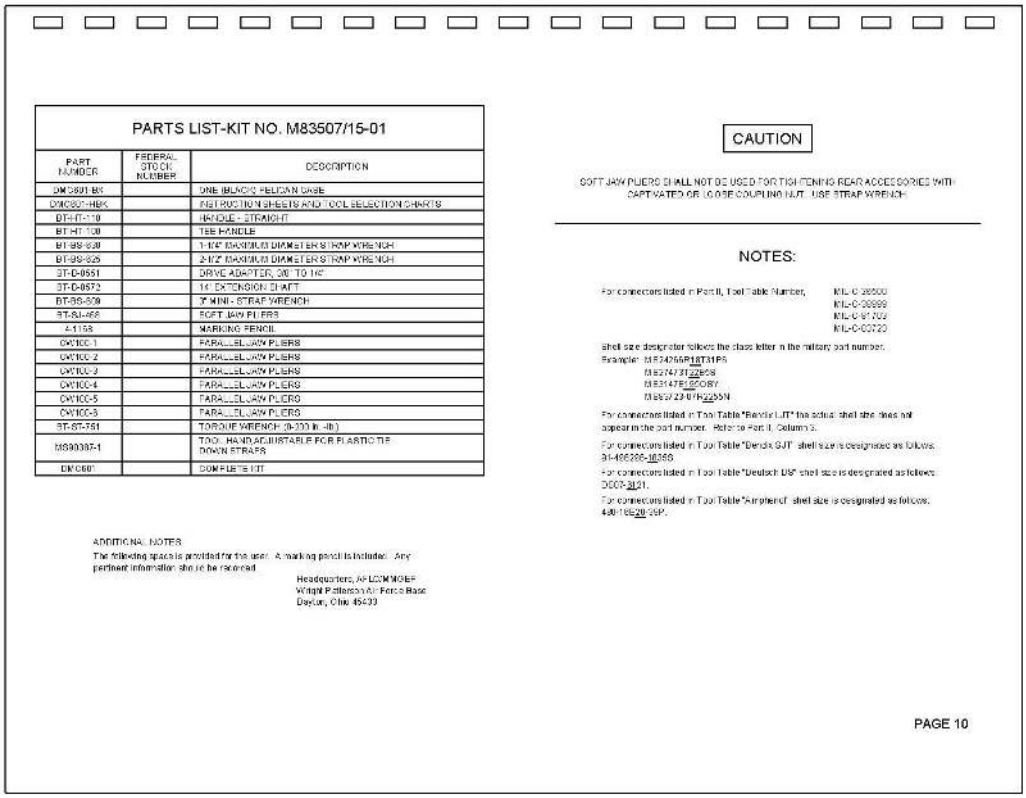


FIGURE 31 Plastic case. Instruction sheet.

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PARTS LIST-KIT NO. M83507/15-01

PART NUMBER	FEDERAL STOCK NUMBER	DESCRIPTION
DM 0601 BK		ONE (BLACK) PELICAN CASE
DMC601-HBK		INSTRUCTION SHEETS AND TOOL SELECTION CHARTS
BT-HT-110		HANDLE - STRAIGHT
BT-HT-100		TEE HANDLE
BT-SS-630		1-1/4" (MAXIMUM DIAMETER STRAP WRENCH
BT-SS-625		2-1/2" (MAXIMUM DIAMETER STRAP WRENCH
ST-D-0551		DRIVE ADAPTER, 3/8" TO 1/4"
ST-D-0572		14" EXTENSION SHAFT
BT-SS-609		3" MINI - STRAP WRENCH
ST-SJ-468		SOFT JAW PLIERS
A-1158		MARKING PENCIL
CW10C-1		PARALLEL JAW PLIERS
CW10C-2		PARALLEL JAW PLIERS
CW10C-3		PARALLEL JAW PLIERS
CW10C-4		PARALLEL JAW PLIERS
CW10C-5		PARALLEL JAW PLIERS
CW10C-6		PARALLEL JAW PLIERS
ST-ST-751		TORQUE WRENCH (0-200 in.-lb.)
MS90387-1		TOOL, HAND ADJUSTABLE FOR PLASTIC TIE DOWN STRAPS
DM 0601		COMPLETE KIT

ADDITIONAL NOTES

The following space is provided for the user. A marking pencil is included. Any pertinent information should be recorded.

Headquarters, AFLCMC/WIEF
Wright Patterson Air Force Base
Dayton, Ohio 45433

CAUTION

SOFT JAW PLIERS SHALL NOT BE USED FOR TIGHTENING REAR ACCESSORIES WITH CAPTIVATED OR LOOSE COUPLING NUT. USE STRAP WRENCH.

NOTES:

For connectors listed in Part II, Tool Table Number, MIL-C-26500
MIL-C-38999
MIL-C-81703
MIL-C-83723

shell size designator follows the class letter in the military part number.
Example: M83726/[illegible]R18T31P5
M83723/[illegible]R22205
M83147/[illegible]60ORY
M83723/[illegible]R22255N

For connectors listed in Tool Table "Bendix GJT" the actual shell size does not appear in the part number. Refer to Part II, Column 3.

For connectors listed in Tool Table "Bendix GJT" shell size is designated as follows: 97-[illegible]3369-16369.

For connectors listed in Tool Table "Deutsch DS" shell size is designated as follows: DEC[illegible]-3121.

For connectors listed in Tool Table "Airpax/Cf" shell size is designated as follows: 480-[illegible]E20-29P.

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FIGURE 32 Plastic case. Instruction sheet.

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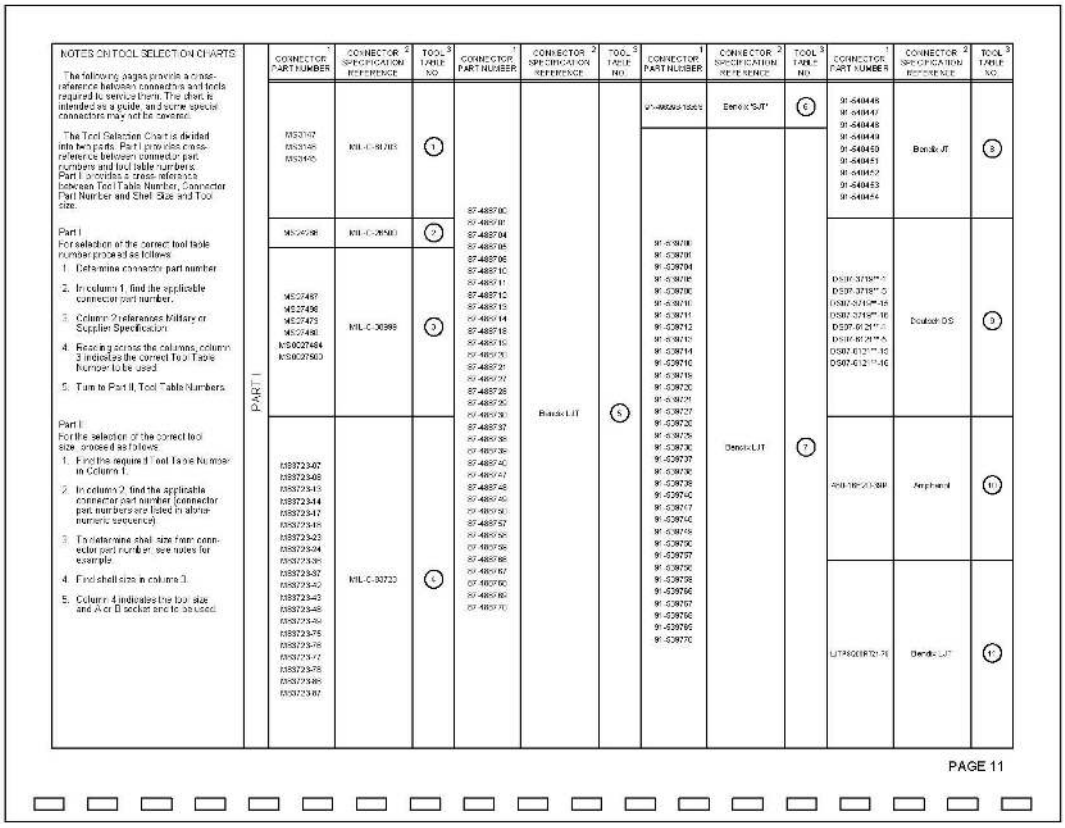


FIGURE 33 Plastic case. Instruction sheet.

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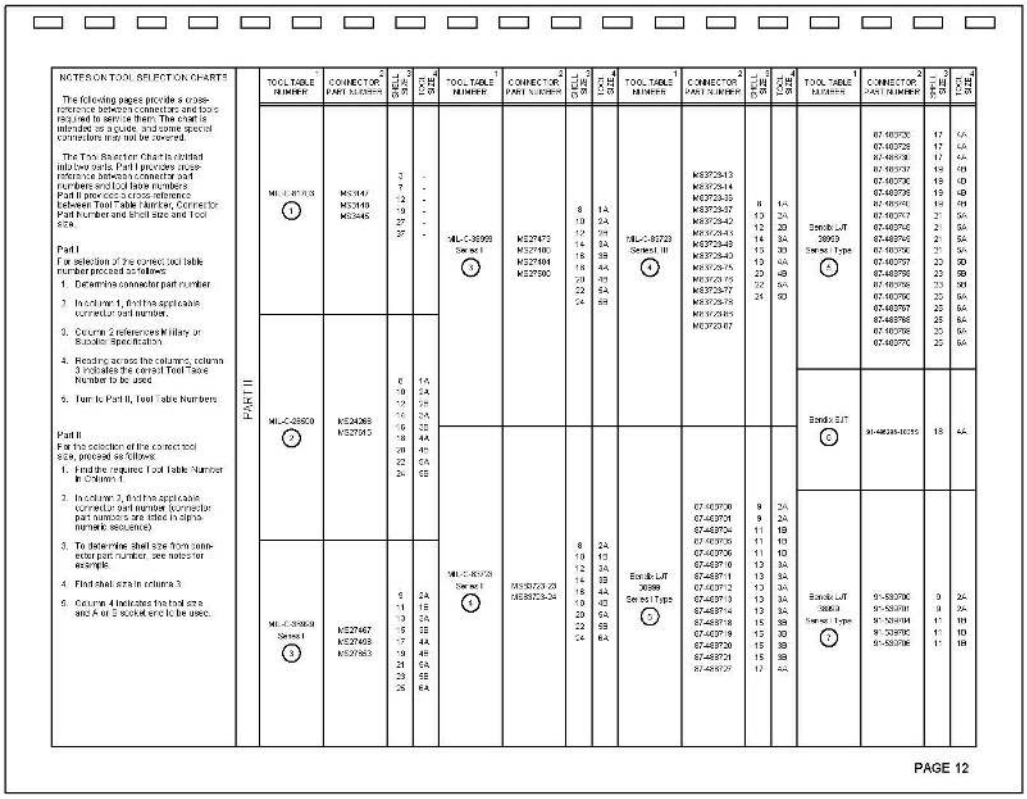


FIGURE 34 Plastic case. Instruction sheet.

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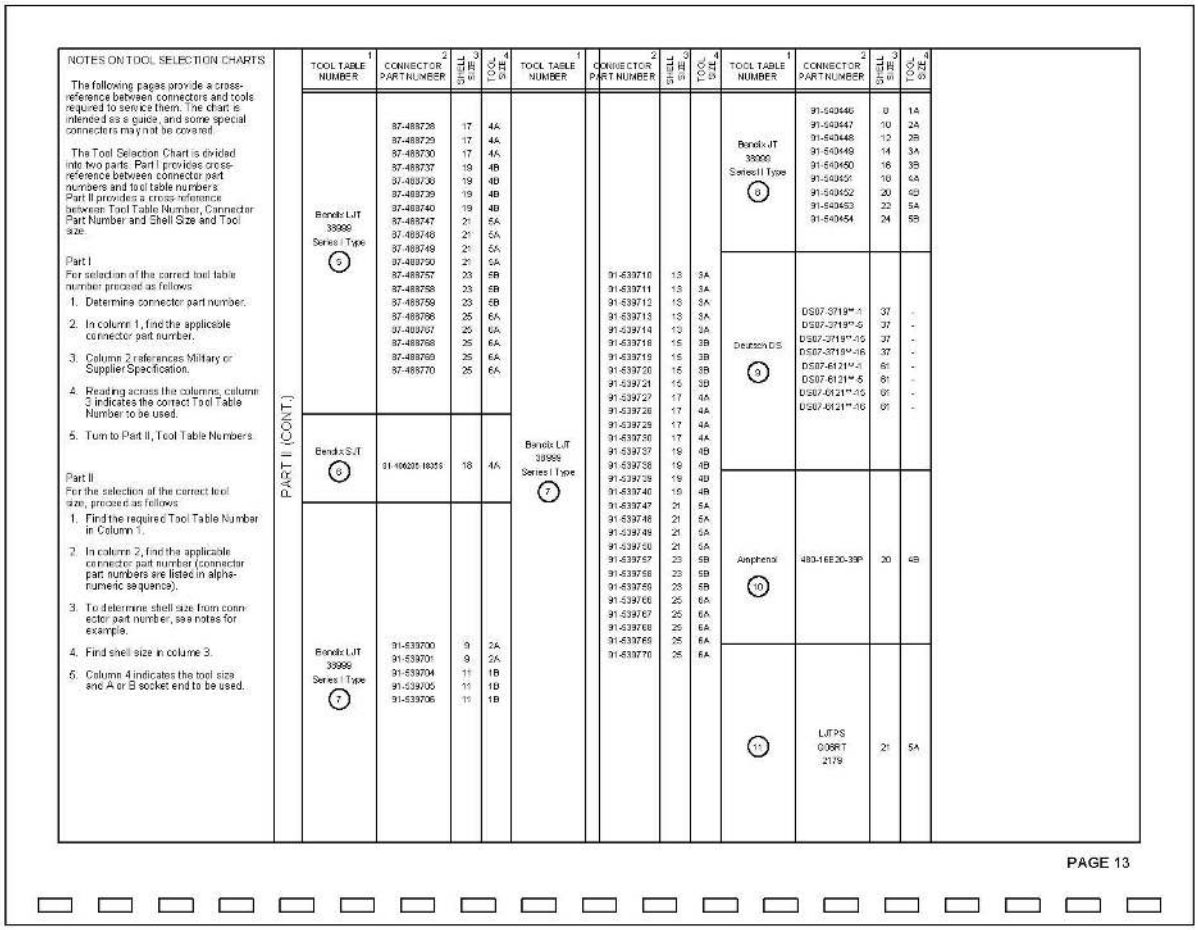


FIGURE 35 Plastic case. Instruction sheet.

REQUIREMENTS:

Materials:

Case (Category 2): fiberglass with an aluminum frame (Skydyne part no. SK99340 or approved equivalent); color olive drab in accordance with SAE-AMS-STD-595.

Cover and components: Aluminum alloy, clear anodized in accordance with MIL-PRF-8625.

Case plastic – Pelican part number 1600

Instruction sheets and contents charts: White vinyl 0.020. inch (0.51 mm) thick.

Insert and retainer pad: Fire retardant foamed polyethylene in accordance with A-A-59135 or A-A-59136.

Clasps, handles, hinges, and locking hasp: Suitably plated to resist corrosion.

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Nameplate: In accordance with MIL-DTL-19834.

Cover: A retainer pad shall be permanently affixed to the underside and a latching mechanism shall be provided so that when the cover is closed the tools will be captivated in their respective positions.

Manual air pressure relief valve: Located adjacent to handle. Decal with "OPEN BEFORE AIRFLIGHT", "CLOSE AFTER AIRFLIGHT" shall be attached to case adjacent to valve.

Rubber feet: Provided on the back and bottom of the case.

Instruction sheets and contents charts: Lettering shall be legible and protected with a transparent coating. Contents charts (centered on inside of top and bottom of tool case cover) shall have pressure sensitive backing and show the insert cavity configurations a:id the respective tool part numbers; a list of NSN will also be supplied.

Part numbers:

M83507/15-01 - Tool kit, general purpose, backshell hardware tools .
M83507/15-02 - Tool case assembly (tools not included)

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83507, this document references the following:

A-A-59135
A-A-59136
MIL-DTL-19834
MIL-PRF-8625
SAE-AMS-STD-595
SAE-AS90387

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy – AS
Air Force - 11

Preparing activity:
DLA - CC

Review activities:
Army – AR, AT. GL MI
Navy - EC, MC, YD
Air Force - 19
DLA - GS

(Project 5180-2025-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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